



Summer Training Report

Tech Elecon Pvt Ltd.

Submitted by

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Enrollment-no : - 12102080701047

In partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

In

Information Technology

Madhuben & Bhanubhai Patel Institute of Technology.

The Charutar Vidya Mandal (CVM) University,

Vallabh Vidyanagar - 388120

April, 2025

Madhuben & Bhanubhai Patel Institute of Technology.

Information Technology



ACKNOWLEDGEMENT

Successful people are that who receive a great deal from their fell people, usually incomparably more than corresponds to their service to them, the value of people, however, should be seen in what they give, and not in what they are able to receive.

-Albert Einstein

It gives me great pleasure in submitting this project entitled “**World Traveller Application**” Developed by our team as a part of the curriculum in IT (Semester VIII). We would like to take opportunity to thank my college, Tech Elecon Pvt Ltd., for giving me this tremendous opportunity to work in the real-time project.

-Thank you very much



**Madhuben and Bhanubhai Patel Institute of Technology
(A Constitue College of CVM University)
New Vallabh Vidyanagar, 388121
Information and Technology**

CERTIFICATE

This is to certify that **Jay VipulBhai Prajapati (12102080701047)** has submitted the Industrial Internship report based on internship undergone at **Tech Elecon Pvt.Ltd.** for a period of **16 weeks from 1 January to 30 april** in partial fulfillment for the degree of **Bachelor of Engineering in Information & Technology , Madhuben and Bhanubhai Patel Institute of Technology** at The Charutar Vidya Mandal (CVM) University, Vallabh Vidyanagar during the academic year 2025 – 2026.

Sign:-

Prof. Jimesh Rana

Sign :-

Prof. Jagruti Prajapati
(HOD)



INTERNSHIP LETTER



24.12.2024

HR/TEPL/145

To,

Madhuben & Bhanubhai Patel Institute of Technology,
V V Nagar.

Permission for Internship

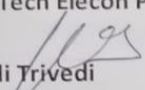
Dear Sir/M'am,

With reference to your letter dated.17.12.2024, we are pleased to grant permission to **Mr.Krish Patel / Mr.Jaimin Shah & Mr.Jay Prajapati** student of **Information Technology for Internship** in our organization from **01.01.2025 to 30.04.2025**.

The following points are to be adhered by the student.

- Laptops, Pen-Drives & other hard drives are not permitted inside the company premises.
- Appropriate dressing & grooming will be appreciated. Jeans & T Shirts are not allowed.
- Female students have to wear dress and Kurtis only.
- Photography & Videography is strictly prohibited in the premises.
- Trainee/s has to strictly comply with the Company's timings & schedule.
- In case of any casualty the Company will not be held responsible.
- On Completion of training the trainee/s has to submit a copy of Training Report duly signed by concern guide to obtain the Training Certificate.
- In case of any ambiguity / difficulty, the trainee is required to approach HR Department. The detailed guideline are attached overleaf.

For, Tech Elecon Pvt.Ltd.


Nirali Trivedi
Group HR Head
Cc. Satyam Raval

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Kolkata: 68, Sarat Bose Road, Kolkata - 700025. Ph. No: 033 24761861 / 24760904



Internship Certificate



HR/TEPL/298

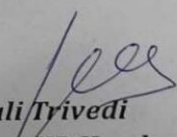
30.04.2025

CERTIFICATE

This is to certify that **Mr. Jay Prajapati** Student of **Information Technology** from **Madhuben & Bhanubhai Patel Institute of Technology - V V Nagar** has undergone for **Internship** in our organization and completed successfully during **01.01.2025 to 30.04.2025**.

During this period, we have found her regular, sincere and enthusiastic.

For, Tech Elecon Pvt. Ltd.


Nirali Trivedi
Group HR Head

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Kolkata: 68, Sarat Bose Road, Kolkata - 700025. Ph. No: 033 24761861 / 24760904



TO WHOM IT MAY CONCERN

This is to certify that Jay VipulBhai Prajapati, a student of Madhuben & Bhanubhai Patel Institute of Technology has successfully completed his internship in the field of Flutter from 1 January to 30 April (16 weeks) under the guidance of Mr. Satyam Raval at TEPL.

His internship activities include successful completion of the assigned project at the given period of time along with abiding by companies' rules and regulations..

During the period of his internship program with us, he had been exposed to different processes and was found diligent, hardworking, and inquisitive.

We wish him every success in his life and career.

For Tech Elecon Pvt Ltd.,



DECLARATION

I, **Jay VipulBhai Prajapati (12102080701047)**, hereby declare that the Industrial Internship report submitted in partial fulfillment for the degree of Bachelor of Engineering in **Information & Technology, Madhuben & Bhanubhai Patel Institute of Technology**, The Charutar Vidya Mandal (CVM) University, Vallabh Vidyanagar, is a bonafide record of work carried out by me at **Tech Elecon Pvt.Ltd.** under the supervision of **prof. Jimesh Rana** and that no part of this report has been directly copied from any students' reports or taken from any other source, without providing due reference.

Name of the Student

Sign of Student



ACKNOWLEDGEMENT

I hereby, would like to have the privilege to show our gratitude to all the persons, helped me whenever I need and provide me the way for the successful completion of this internship without hindrance. I am grateful to all our mentors who inspired me by setting an example of them for the kind purpose of motivating me to reach my targeted objective. Without their knowledge and wisdom along with experience and specialization in their specific field, I would not have been able to think of doing or completing this work. All the persons who have contributed directly or indirectly with their kind support and humble approach are highly appreciative and I would always remain indebted to them in all the ways.

I would like to take the opportunity to thank and express my deep sense of gratitude corporate mentor Mr. Satyam Raval (Deputy General Manager) ,Mr. Chirag Maheta. I am really grateful to them for providing their valuable guidance at all stage of the internship, their advice, constructive suggestions, positivity and supportive attitude and continuous encouragement, and innovation has not only felicitied my learning but also allowed me to explore my potential and contribute meaningfully to the project without it would have not been possible to complete the internship.

I would also like to extend my gratitude to our internal guide prof. Jimesh Rana Sir, helped me with all the practical insights and aspects of my project also whenever the guidance needed sir was always ready to share his experience and advice.

I am thankful to Tech Elecon Pvt. Ltd. for giving me the opportunity to develop my skills and grow myself



ABSTRACT



I have done my internship at Tech Elecon Pvt. Ltd. Anand – Gujarat as a Junior Flutter Developer intern, the main purpose of the internship was to fulfill the core equipment for the award of bachelor's degree in Information & Technology and gain practical experience in the IT industry.

Throughout the Internship I have worked on programming languages like Flutter, php, node, mongo , sql also with this I've done my internal project named "CCNTC" and "Shayona Automob". This project outlines the role of a Flutter Developer Intern specializing in Flutter. The internship aims to provide hands-on experience and practical knowledge to developers in the field of Mobile App development using Flutter.

"CCNTV" is an OTT platform based Tv and Mobile which includes learning and applying Flutter Dart programming concepts, understanding database integration, and collaborating with the development team to deliver high-quality software solutions. Since Flutter is an open source Frame work with Dart programming language, we've worked under the guidance of experienced developers, contributing to the design, development, and maintenance of Mobile applications.

Overall, this internship provides a valuable learning experience for individuals seeking to start their careers as Mobile App developers, offering mentorship, real-world projects, and the chance to contribute meaningfully to the development of innovative web applications.



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List of Abbreviations & Symbols



Abbreviations

IT
UI
Flutter
XAMPP

WF
Dart
RFID
KPI
DFD
API
QA
KPI

Full Form

Information Technology
User Interface
FrameWork for Mobile app Developement
Cross-platform(X), Apache (A), MySQL
(M), PHP (P), Perl (P)
Work Flow
Programming Language
Radio Frequency Identification
Key Performance Indicator
Data Flow Diagram
Application Programming Interface
Quality Assurance
Key Performance Indicator



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CHAPTER 1

OVERVIEW OF COMPANY

We want to transform businesses with innovative solutions. We're here to redefine how things work in our industry, creating smooth digital experiences that make a real impact. We're all about excellence, building long-lasting partnerships, and exploring endless possibilities for a future full of new ideas.

Tech Elecon PVT Ltd, a certified Company established in 2002. We're a dedicated team of 150+ passionate, creative, and innovative IT experts committed to turning digital ideas into reality. In addition to our certification, we proudly offer diverse IT services and domain-specific solutions. Our devoted team is committed to delivering the best industry practices, combining technology expertise with business domain knowledge for effective digital transformation.

Recognizing the uniqueness of each business, we take the time to understand clients' needs and craft personalized solutions from ideation to launch and beyond. We take pride in delivering projects on time, within budget, and to the highest quality standards. Staying ahead of the curve, we continuously research and implement the latest technology and trends. Whether it's a new website, a mobile app, or ongoing maintenance and support, TechElecon (TEPL) is here to make it happen.

Moreover, we provide flexible hiring options, allowing you to hire dedicated developers for your projects on an hourly, part-time, or full-time basis. This ensures you have the flexibility and scalability needed to meet your business requirements effectively.

CHAPTER 2

COMPANY PROFILE

2.1 Company Profile

Tech Elecon Pvt. Ltd. is the It division of the Elecon group of companies and has more than 25 years of experience in the fields of hardware, software, and networking solutions.

It is situated in the heart of Vithal Udyognagar Industrial Estate and in the proximity of the educational town of Vallabh Vidyanagar.

Tech Elecon all set to reach new heights in the field of IT solutions. Tech Elecon is ready with all sorts of solutions and delivers any application that is web based and further our solutions are designed to adapt your business rather than your business adapting the software. Their solutions are 100% fruitful and empower you to take control of client's business online and in real time.

Tech Elecon have more than 100 employees with specialized skills in software development, custom software development, and e-commerce software development using custom software programming including .NET, C#.NET, PHP, and Open Source and Oracle.

Tech Elecon delivers quality products and services with a focus on integrating the same with existing technologies, providing the required automation to our customers to help them achieve their business objectives.

Mr. Nilesh Naik, the company's Vice President, is at the helm of the Tech Elecon organization. Mr. Satyam Raval, as Deputy General Manager, and after that, Manager and Associate Manager positions are listed. At the bottom, there are trainees at entry level, who follow up to engineer, senior engineer, also executive and senior executive manager.

2.2 Different Departments of Company

Tech Elecon has extensive experience in providing IT services and has successfully adapted to technological advancements, making it the leading IT infrastructure management service provider in the region. Our cutting edge delivery model covers all the stages of the solution lifecycle, including planning, deploying, managing, maintaining, auditing, upgrading, and improving.

Tech Elecon recognize that each client has unique needs and expectations when it comes to infrastructure and service providers. Our clients have the flexibility to choose from a wide range of IT infrastructure management and performance services based on their specific requirements. They can opt for on-site services on hybrid solutions that include on-site troubleshooting and support services.

Tech Elecon provides various services for business:

- Hardware maintenance and repairing
- Service desk management
- Desktop management
- Network management
- Messaging administrator
- Back-up managemnent

Other services:

- Software Development Services
- Software Licensing
- Microsoft Product Implementation
- Linux Servers/ Desktop Implementation

CHAPTER 3

INTRODUCTION TO INTERNSHIP

3.1 Internship Summary

Internship Position: Flutter & React Developer Intern (Flutter Department Intern Head)

Company: Tech Elecon

Duration: 16 Weeks

Domains: Flutter & React Development

Internship Overview:

During my 16-week internship at Tech Elecon, I had the opportunity to work across two key development domains—Flutter and React. This dual-role internship allowed me to gain valuable hands-on experience in mobile and web application development. Over the course of the internship, I successfully developed and delivered five major projects: three in Flutter and two in React. I actively participated in all phases of the development lifecycle, including requirement analysis, UI/UX design, coding, testing, and deployment.

In addition to my development responsibilities, I served as the **Flutter Department Intern Head**, leading and mentoring fellow interns in the Flutter team, organizing peer code reviews, and ensuring smooth coordination between UI/UX designers and developers.

3.2 Roles and Responsibilities

As a Flutter Developer Intern & Team Head:

- Led the Flutter intern team, coordinated task allocation, provided mentorship, and ensured timely project delivery.
- Designed and developed mobile applications with modern, user-friendly UIs across multiple platforms.
- Integrated RESTful APIs for real-time data fetching and dynamic content display.
- Conducted extensive testing and debugging to maintain performance and compatibility across Android and iOS devices.
- Participated in agile sprint planning, daily stand-ups, and retrospective meetings.
- Collaborated closely with UI/UX designers and backend developers to ensure a seamless development pipeline.

As a React Developer Intern:

- Developed scalable and responsive web applications using React.js.
- Worked on frontend logic, component-based architecture, and efficient state management.
- Collaborated with backend developers to ensure proper API integration and error handling.
- Conducted UI optimization and improved web performance for user satisfaction.

3.3 Projects Undertaken

Flutter Projects

1. OTT Platform

- **Objective:** Build a mobile streaming app that allows users to browse, watch, and manage video content.
- **Key Features:**
 - Video listing by categories (Movies, Web Series, Trending)
 - Video player integration and UI controls
 - User authentication and personalized recommendations
- **Technologies Used:** Flutter, Dart, Firebase, node mongo API, 3 Servers, Different Security Utils.

2. Pregmed – Pregnancy Tracker App

- **Objective:** Help expecting mothers track and manage their pregnancy journey week-by-week.
- **Key Features:**
 - Weekly health tips and progress tracking.
 - Appointment reminders and baby growth visualization.
 - Notification alerts and data privacy
- **Technologies Used:** Flutter, Firebase, Dart

3. QR Code-Based Attendance Management System

- **Objective:** Automate employee or student attendance using QR scanning.
- **Key Features:**
 - Unique QR code generation and scanning
 - Admin dashboard for attendance reports
 - Real-time Firebase data storage
- **Technologies Used:** Flutter, Firebase Realtime Database, Dart, QR Code Package

React Projects

1. CRMS – Customer Relationship Management System.

- **Objective:** Manage customer information, interactions, and data insights.
- **Key Features:**
 - Contact management, lead tracking, and support tickets.
 - Interactive dashboard with analytics.
 - Role-based access control.
- **Technologies Used:** React.js, Axios, Node.js (for backend), MongoDB.

2. Billing Management System

- **Objective:** Automate and simplify billing operations for small businesses.
- **Key Features:**
 - Invoice generation and billing record keeping.
 - Product and inventory management.
 - PDF invoice export and email integration.
- **Technologies Used:** React.js, Express.js, MongoDB

3.4 Tools & Technologies Used

- **Flutter & Dart** – For building cross-platform mobile applications.
- **React.js** – For developing dynamic and responsive web applications.
- **Firebase** – For authentication, data storage, and push notifications.
- **MongoDB, Node.js, Express.js** – Backend and database for React projects.
- **VS Code, Android Studio** – Development IDEs,
- **Git & GitHub** – Version control and collaboration.

CHAPTER 4: SYSTEM ANALYSIS

4.1 Introduction

This chapter provides a comprehensive analysis of the key projects undertaken during my 16-week internship at Tech Elecon, including the OTT Platform (CCNTV), Shayona Automobile Management System, Pregmed Pregnancy Tracker App, QR Code-Based Attendance Management System, CRMS, and Billing Management System. The analysis covers objectives, scope, system architecture, and implementation strategies through literature reviews, system design, and feasibility assessments.

1. Literature Review

OTT Platform (CCNTV):

- Analyzed existing OTT platforms such as Netflix, Amazon Prime, and Hotstar to understand streaming technology, subscription models, and content delivery mechanisms.
- Explored challenges in video streaming, including latency, bandwidth optimization, and DRM (Digital Rights Management) for content security.
- Evaluated best practices in UI/UX for smart TVs, mobile apps, and web-based streaming services.

Shayona Automobile Management System:

- Studied traditional garage and spare parts management systems to identify inefficiencies in service tracking, inventory handling, and customer management.
- Reviewed available automobile ERP solutions to understand data flow, invoice generation, and service history tracking.
- Identified challenges in integrating cloud-based solutions with offline businesses such as garages and spare parts retailers.

Pregmed – Pregnancy Tracker App:

- Studied health apps like BabyCenter and Ovia to understand weekly health content and notification systems.
- Evaluated data privacy concerns and UI/UX practices for healthcare-related apps.

QR Code-Based Attendance Management System:

- Reviewed traditional attendance systems and their shortcomings.
- Researched QR-based technologies for secure and efficient attendance logging.

CRMS – Customer Relationship Management System:

- Analyzed leading CRMs (Zoho, Salesforce) for features like lead tracking, ticketing, and dashboards.
- Studied API integrations and analytics dashboards for customer insights.

Billing Management System:

- Researched invoice generation tools for small businesses.
- Reviewed inventory and tax calculation logic used in standard billing software.

2. System Analysis**OTT Platform (CCNTV):**

Architecture: Server-based with microservices for streaming and payment services.

Features: Video categorization, subscription plans, content security.

Stakeholders: Admin, Content Creators, Users, Support Team.

Requirements:

- Functional: User auth, streaming, payments.

- Non-functional: Scalability, latency, cross-platform compatibility.

Shayona Automobile Management System:

Architecture: Node.js backend with PostgreSQL, deployed on the cloud.

Features: Inventory, employee, vehicle service, and financial management.

Stakeholders: Admin, Garage Owners, Vehicle Owners

Requirements:

- **Functional:** Order, customer, inventory management.
- **Non-functional:** Security, scalability, usability.

Pregmed:

Architecture: Firebase backend with Flutter frontend.

Features: Weekly health tips, appointment reminders, notifications.

Requirements:

- **Functional:** User login, data storage, alerts.
- **Non-functional:** Privacy, accessibility.

Billing Website (React.js + MySQL)

Architecture: React frontend with REST APIs connecting to MySQL database.

Features:

- Item Master for product management.
- Invoice generation with tax calculations.
- Bill preview and export as PDF.

Stakeholders:

Admin/User: Add/edit items, generate bills, view transactions.

QR Code-Based Attendance Management:

Architecture: Flutter app with Firebase Realtime Database.

Features: QR generation, admin-only scanning, live data updates.

Requirements:

- **Functional:** Attendance records, QR scans.
- **Non-functional:** Admin restriction, reliability.

CRMS:

Architecture: MERN stack (MongoDB, Express, React, Node).

Features: Lead tracking, support tickets, analytics dashboard.

Requirements:

- **Functional:** Customer DB, contact logs.
- **Non-functional:** Role-based access, scalability.

3. System Design

OTT Platform:

- Design: Microservices and responsive UI for all devices.
- Database: Users, content, subscriptions, transactions.
- Design Choices: ABR streaming, OAuth2.0, CDN integration.

Shayona Automobile Management:

- Design: Modular system with a scalable schema.
- DB: Inventory, employees, services, financials.
- Design Choices: Role-based access, indexed queries.

Pregmed:

- Design: Modular screens by pregnancy week.
- DB: User profiles, health data.
- Design Choices: Alert notifications, health content modules.

QR Attendance:

- Design: Secure QR generator and scanner.
- DB: Attendance logs per user per day.
- Design Choices: Admin-only scanner, Firebase security rules.

CRMS:

- Design: Interactive dashboard, role-based UI.
- DB: Customers, interactions, leads.
- Design Choices: Axios integration, real-time updates.

Billing System:

- Design: Grid-based itemized billing UI.
- DB: Items, invoices, customers.
- Design Choices: PDF export, responsive layout.

4. System Implementation**OTT Platform:**

- Server setup, payment gateway integration.
- Challenges: Bandwidth -> Used CDN; Piracy -> DRM encryption.

Shayona:

- Node.js backend with real-time syncing.
- Challenges: Spare parts volume -> Indexed queries; Security -> RBAC.

Pregmed:

- Firebase push notifications, weekly data management.
- Challenges: Timely reminders -> Scheduler-based alert system.

QR Attendance:

- Real-time attendance dashboard.
- Challenges: QR spoofing -> One-time QR, admin-only scan.

CRMS:

- Real-time support tracking and CRM dashboard.
- Challenges: Lead duplication -> Unique lead tracking logic.

Billing:

- PDF generation, financial report summary.
- Challenges: Invoice formatting -> PDF libraries for styling.

4.2 New System Requirements

Functional Requirements:

- User Management, Data Tracking, Billing, Role-Based Access across all systems.
- Domain-specific needs: Video playback (OTT), Service tracking (Shayona), Weekly content (Pregmed), QR login (Attendance), CRM features, and Billing modules.

Non-Functional Requirements:

- High performance, scalability, security (OAuth2.0, DRM, RBAC), cross-platform compatibility, usability for both tech and non-tech users.

4.3 System Feasibility

OTT Platform:

- Technically feasible with cloud-hosted, CDN-integrated streaming.
- Operationally smooth with CMS and intuitive UI.

Shayona:

- Technically strong with Node-PostgreSQL stack.
- Operational ease with automated restocking and cloud access.

Pregmed:

- Feasible with Firebase-backed modular design.
- Easy for users with guided content.

QR Attendance:

- Technically robust for real-time scans.
- Operationally controlled with admin-only access.

CRMS:

- Technically capable via MERN stack.
- Operationally useful for sales/service teams.

Billing:

- Feasible for small businesses needing simple billing.
- Operationally effective with responsive dashboards and multi-payment support.

CHAPTER 5: PROJECT MODULES & SYSTEM FLOW DESIGN

5.1 PROJECT MODULES

1. OTT Platform (CCNTV)

a. Design

i. Platform Overview:

- Built using Flutter, the platform ensures seamless performance across Android and iOS devices.
- Integrated with live and on-demand video streaming via real-time APIs and CDN.
- Backend hosted on a cloud-based server for high availability.

ii. User Interface:

- Home Screen: Highlights trending, latest, and genre-based content.
- User Profiles: Multi-profile support with personalized recommendations.
- Playback Screen: ABR streaming with subtitle and dubbing support.

iii. Advanced Features:

- Adaptive Bitrate Streaming (ABR)
- DRM-protected content
- OAuth 2.0 authentication
- Subscription management with auto-renewal and in-app payments.

2. Billing System (React + PHP + MySQL)

a. Design

i. Web Dashboard:

- Built with React.js, using a modular component structure.
- Backend API developed in **PHP**, connected to a **MySQL** database.
- Real-time CRUD operations for invoices and inventory.

ii. Key Interfaces:

- **Login/Register:** Role-based login (admin, cashier, viewer).
- **Items Master:** Add/edit/delete product information.
- **Billing Form:** Create invoices with tax (GST/VAT) calculations.
- **Bill Preview:** Export bill in **PDF format**.
- **User Profile:** View and edit user info and access history.

iii. Features:

- Real-time billing and stock updates
- Auto tax calculation and total summarization
- Export options (PDF/Print)
- Financial reports with filters and summaries.

3. PregMed (Healthcare App)**a.Design****i. Mobile Application:**

- Built using Flutter and integrated with Firebase Authentication and Firestore.
- Designed to assist expecting mothers by tracking pregnancy stages, doctor appointments, and health tips.

ii. User Interface:

- **Dashboard:** Overview of pregnancy stage, due date countdown, and health tips.
- **Appointment Scheduler:** Allows users to book, view, and cancel doctor visits.
- **Health Journal:** Enables users to record symptoms and notes.

iii. Firebase Features:

- Realtime data sync using Cloud Firestore.
- Firebase Cloud Messaging (FCM) for reminders and notifications.
- Role-based access: users (patients), doctors, and admin..

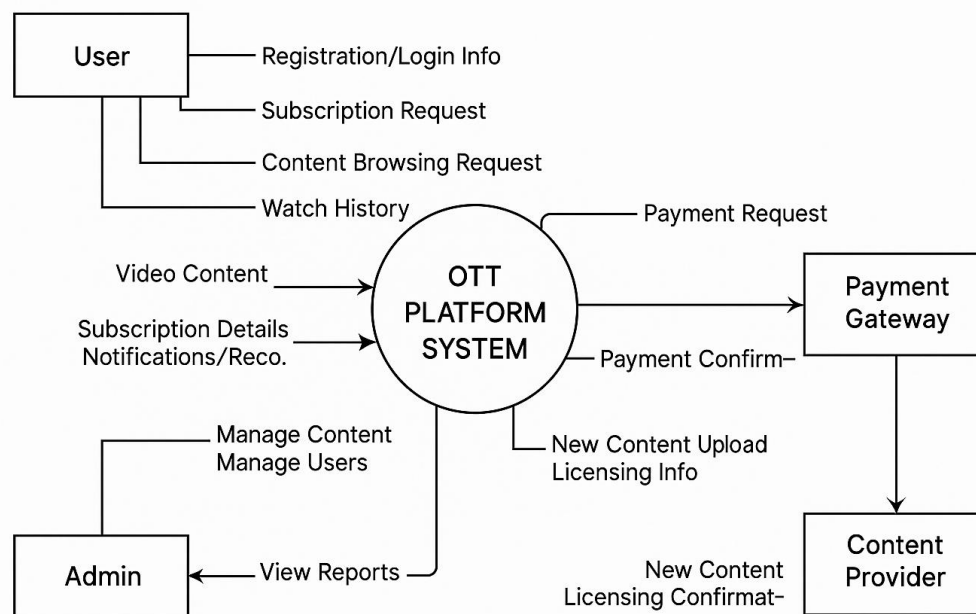
5.2 SYSTEM FLOW DESIGN

1. OTT Platform – Data Flow Design

0-Level DFD

- Users interact with the system through a frontend app.
- Requests for video/content go to the server, which returns video data from the Content DB.
- Admin uploads content and manages subscriptions.

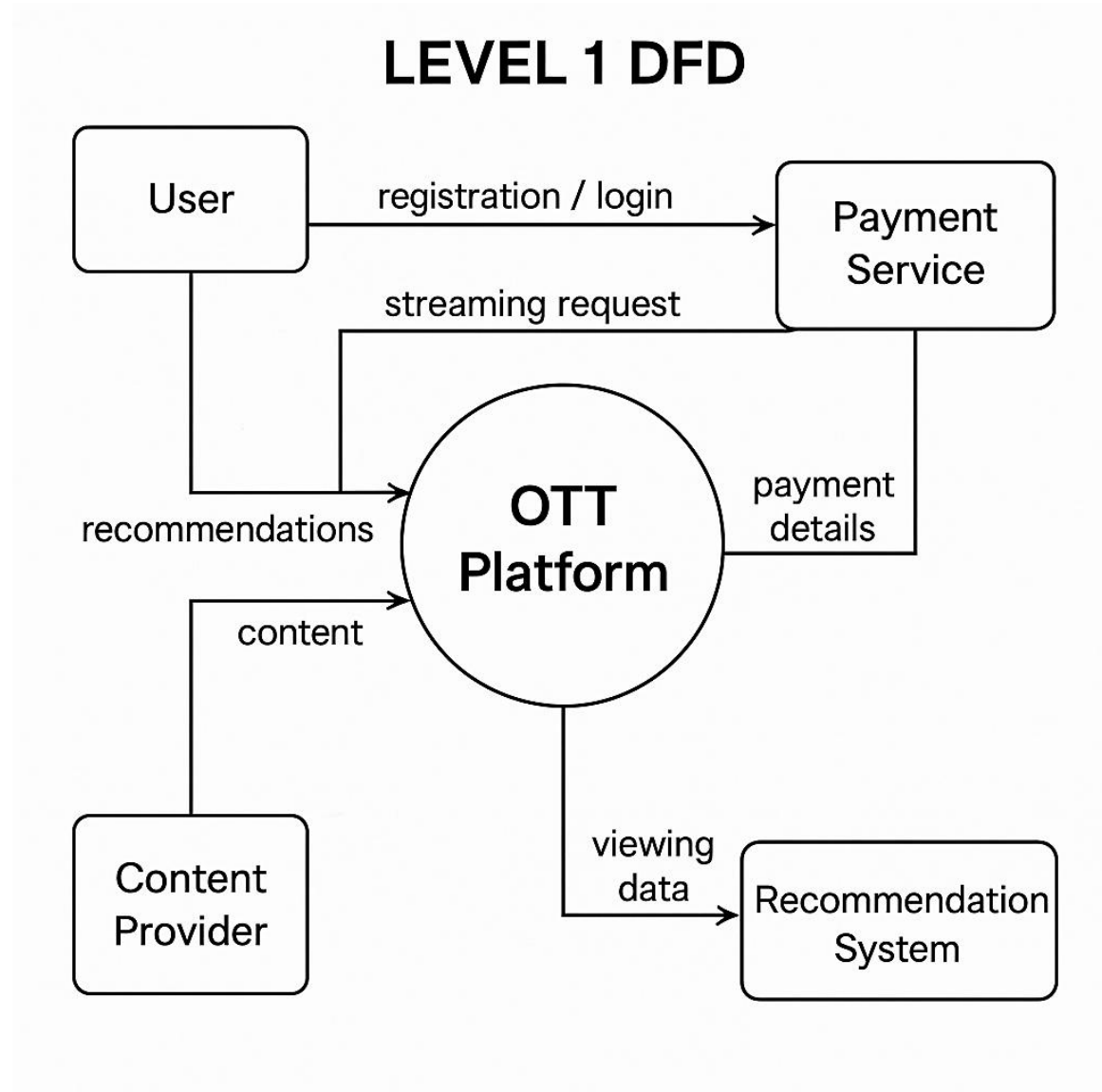
0-Level DFD for OTT Platform



5.1 0-level DFD for OTT Platform

1-Level DFD: User Login & Streaming

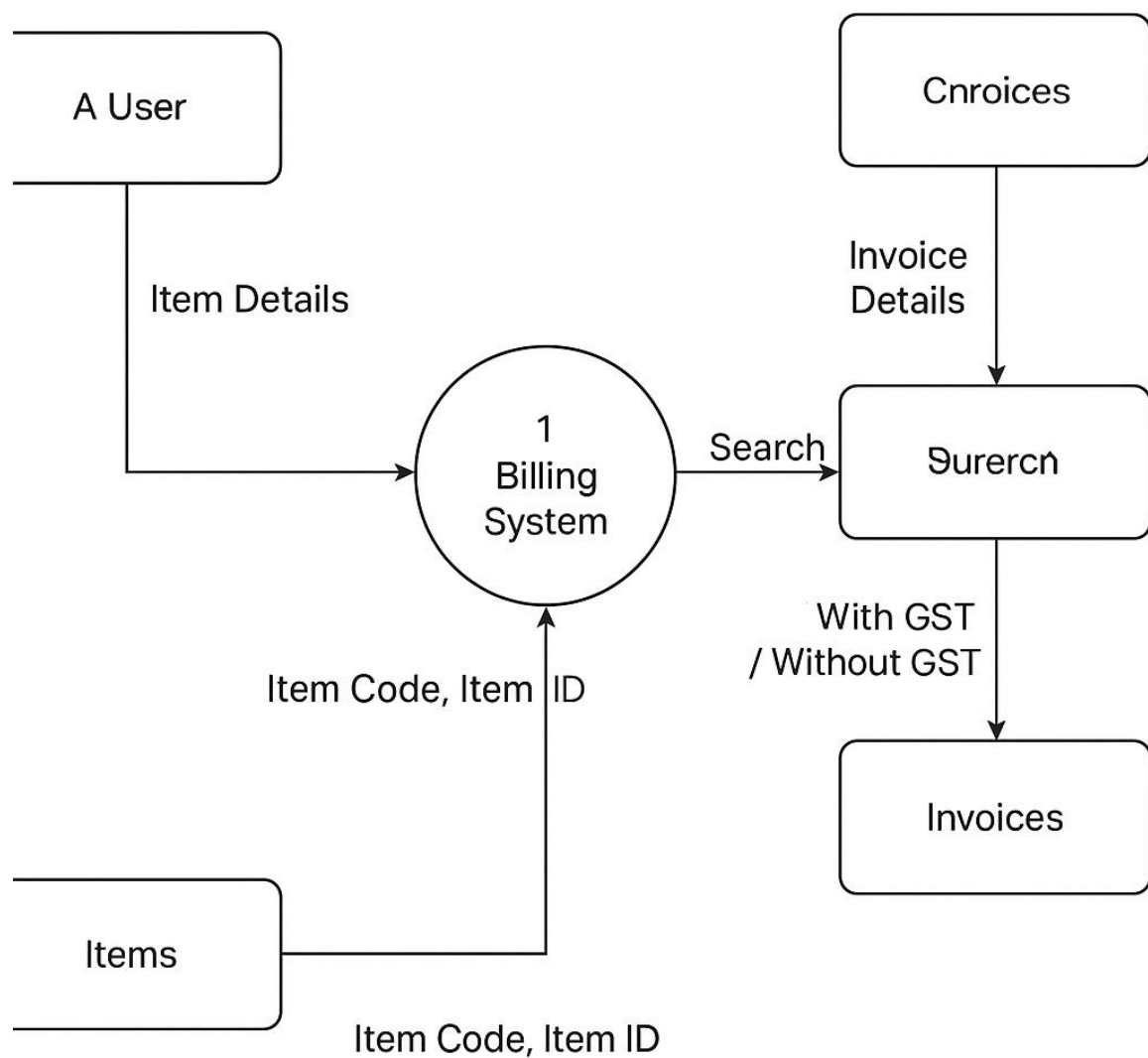
- User inputs login credentials → verified via server
- Upon success, session starts → User selects video
- Server fetches video details → Streams via CDN
- Session ends → Auto logout after timeout/inactivity.

**5.2 Level 1 DFD for OTT Platform**

2. Billing System – Data Flow Design

0-Level DFD

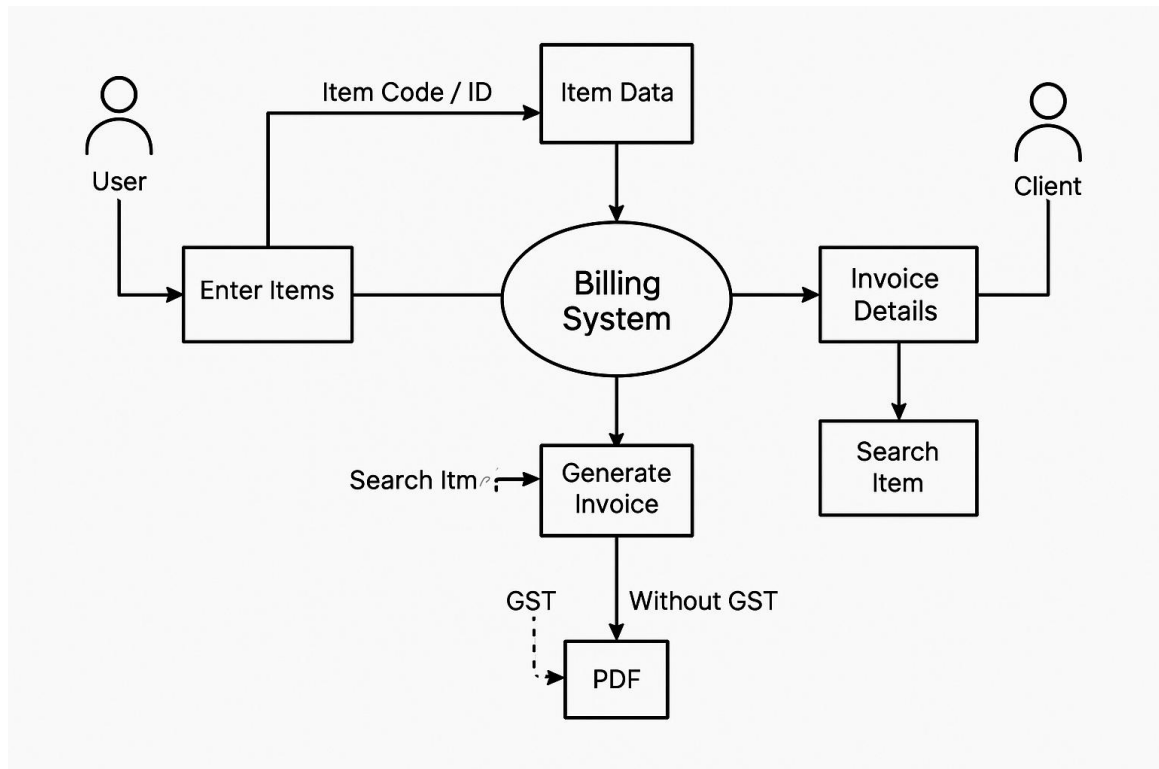
- Admin interacts with the React frontend
- Operations such as product management, billing, and reports go to the PHP API
- API interacts with the MySQL DB



5.3 Level 0 Billing System DFD Diagram

1-Level DFD: Billing Module

- User selects items → Inputs customer details
- System calculates tax, generates invoice
- Invoice saved to DB → Shown on preview
- User exports/prints bill.



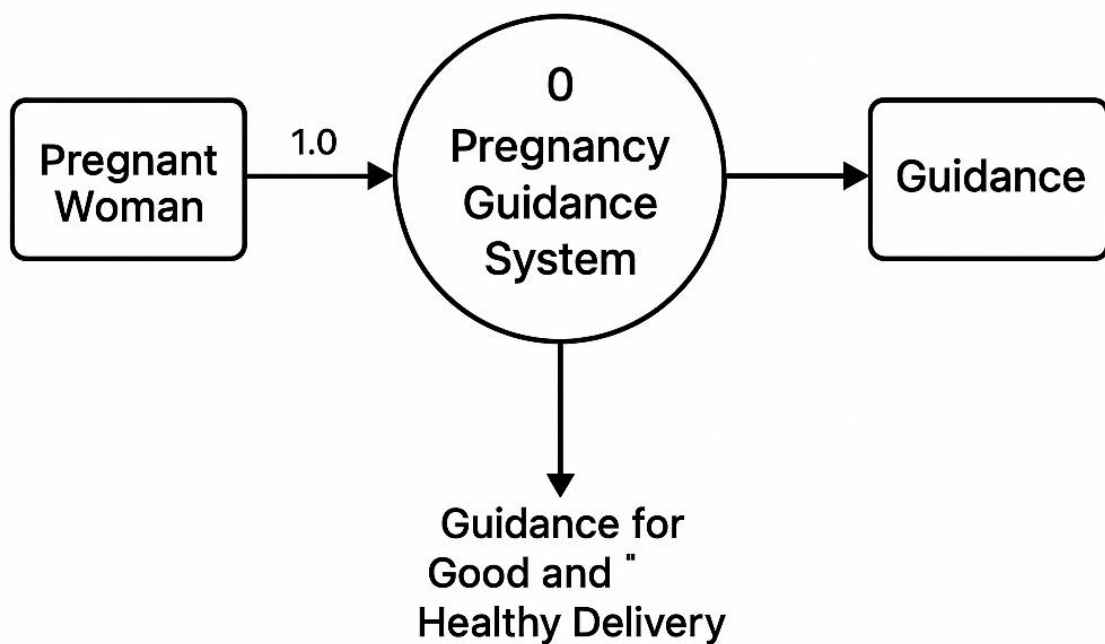
5.4 Level 1 Billing System DFD Diagram

3. PregMed – Data Flow Design

0-Level DFD

- User accesses app → Authenticates via Firebase Auth
- Appointments, symptoms, and notes are stored in Firestore
- Doctors access and update schedules and feedback

Level 0 Data Flow Diagram

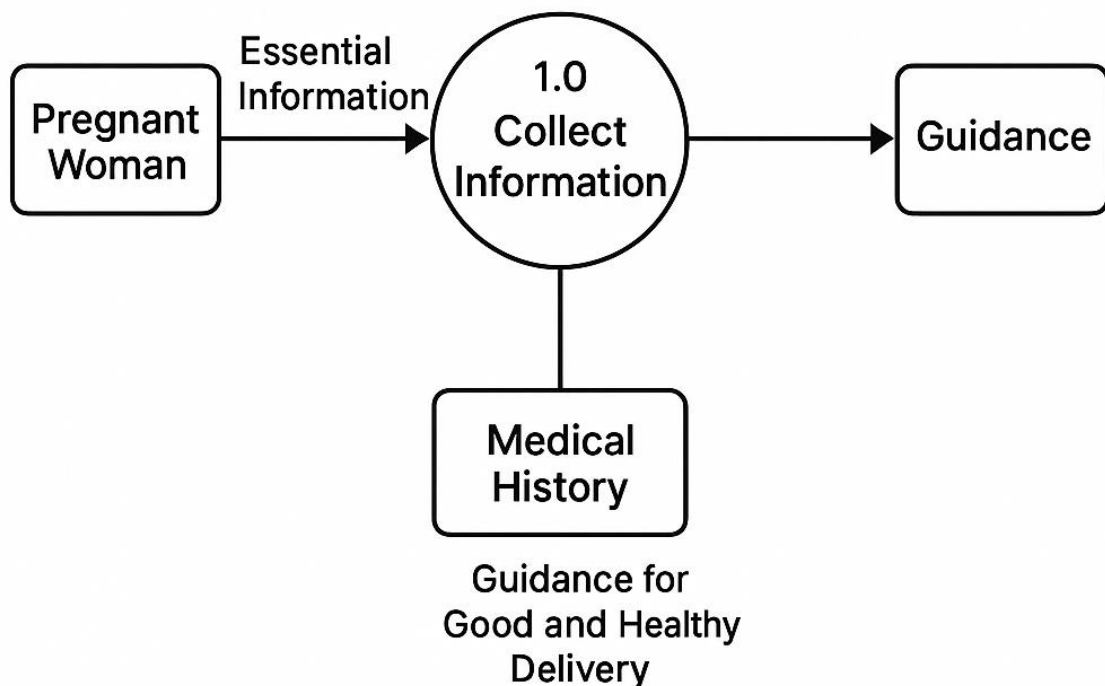


5.5 Level 0 DFD Diagram for Pregmed

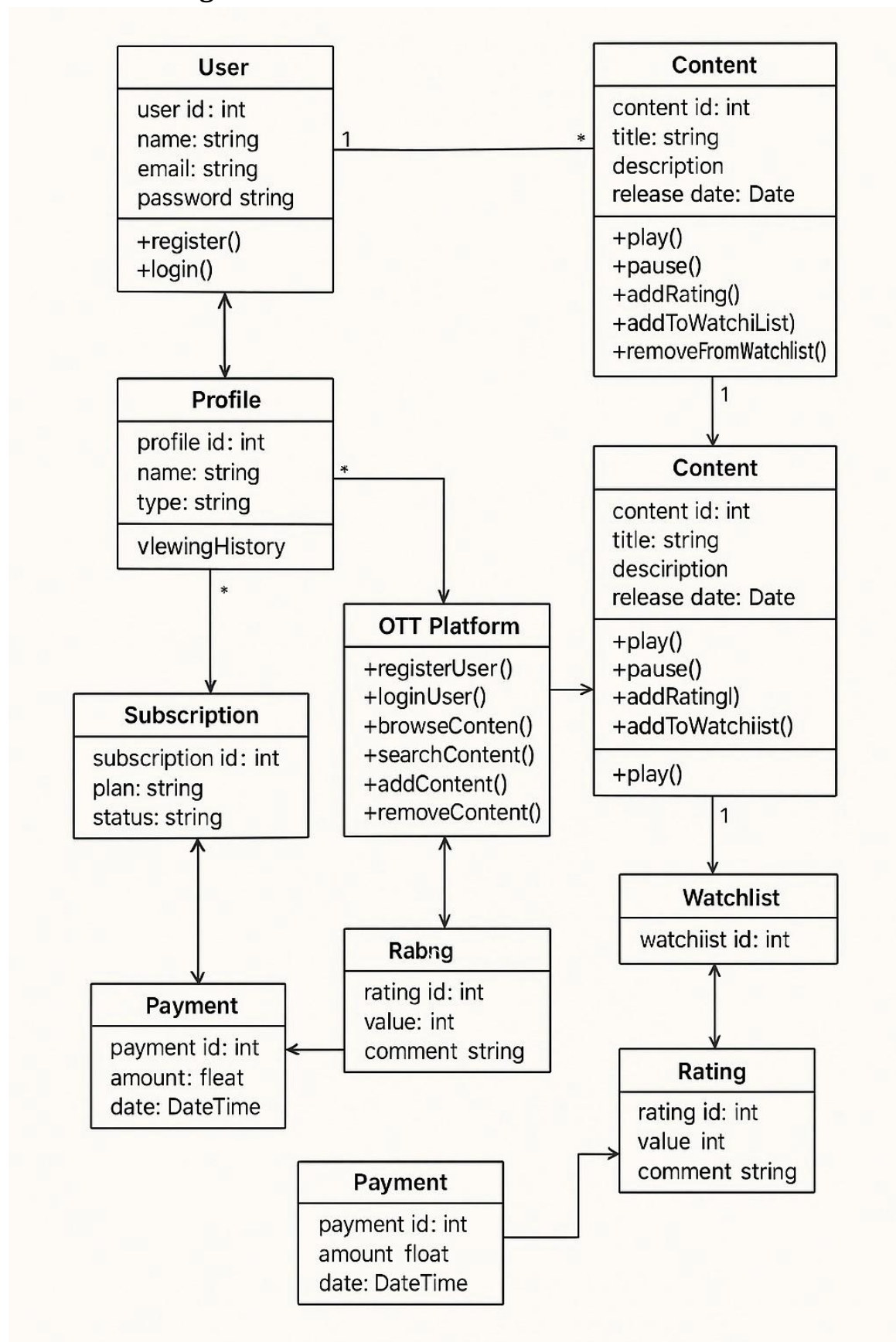
1-Level DFD: Appointment Module

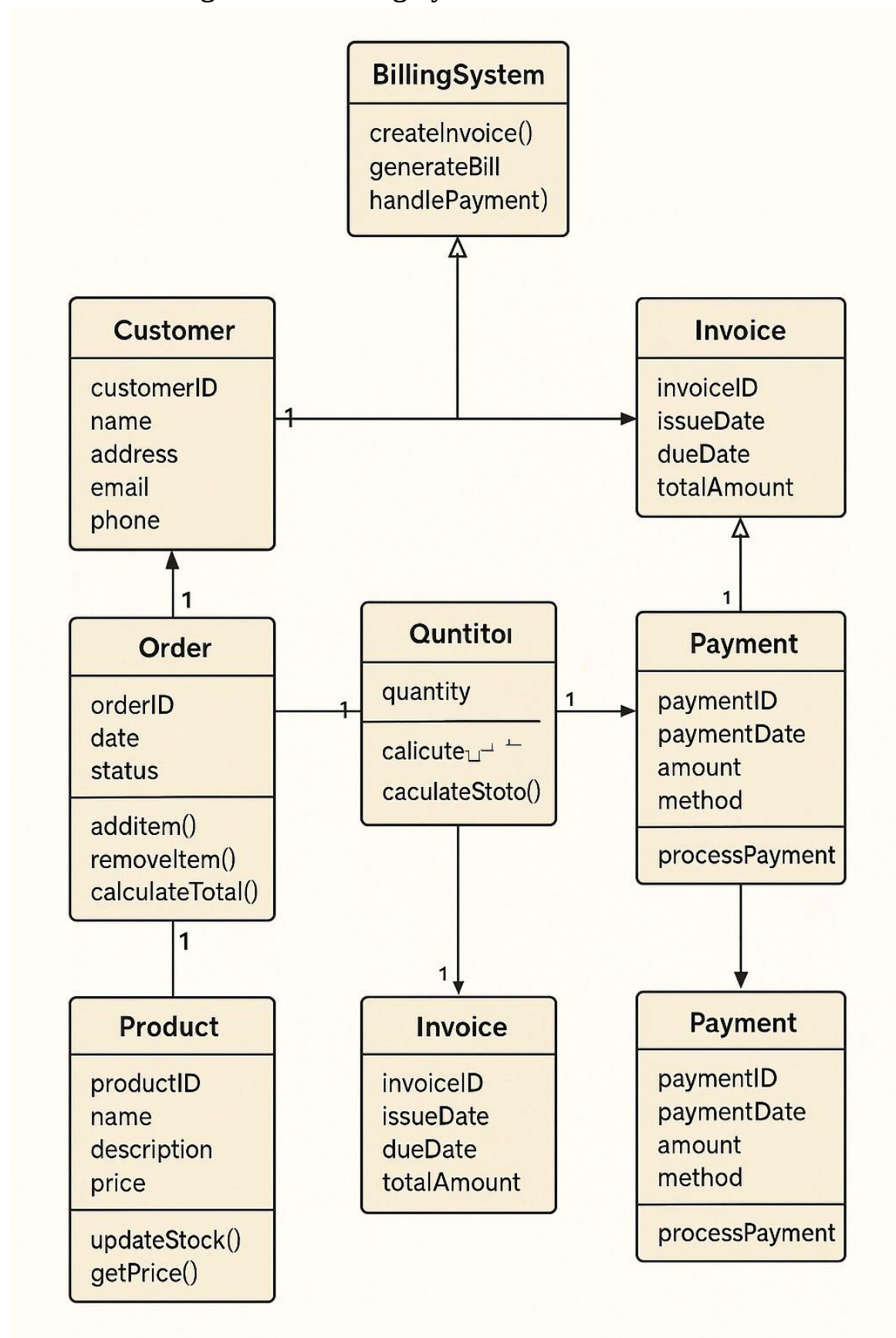
- User selects a date → Chooses doctor/time
- Request sent to Firestore → Check availability
- On success, appointment is saved → Notification sent via FCM.

Level 1 Data Flow Diagram

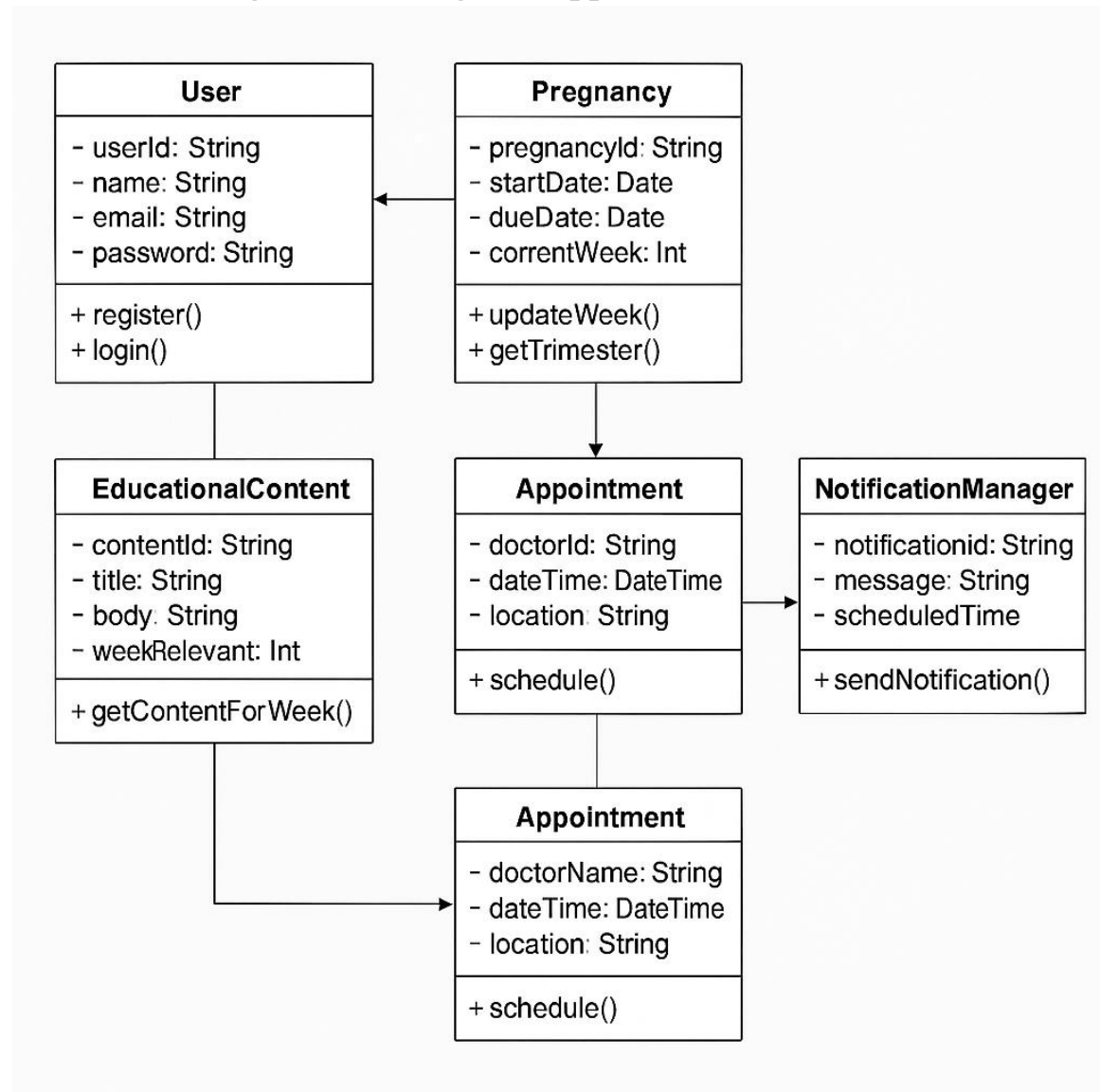


5.6 Level 1 DFD Diagram for Pregmed

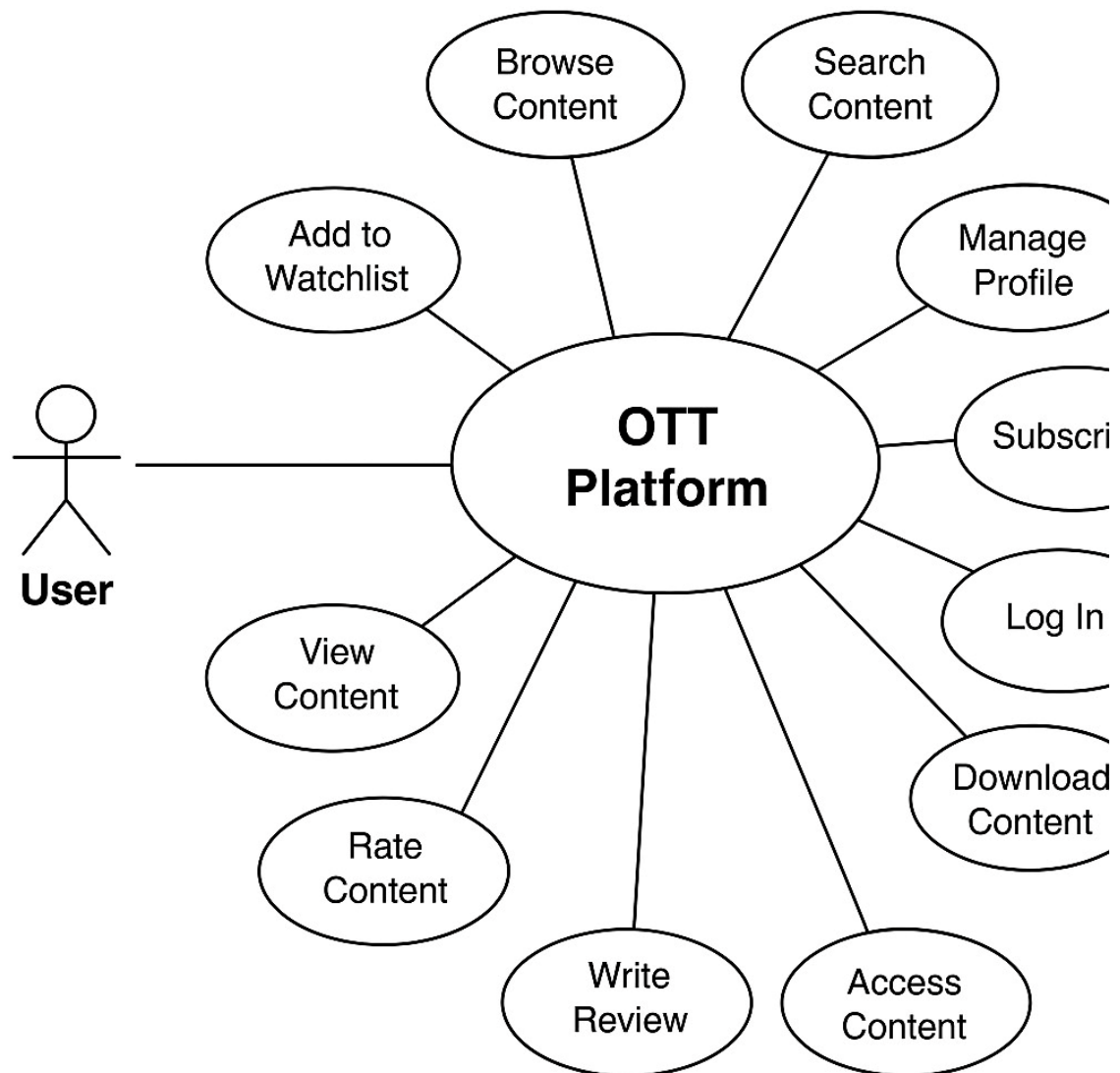
UML Class Diagram For OTT Platform.**5.7 UML Class Diagram for OTT Platform**

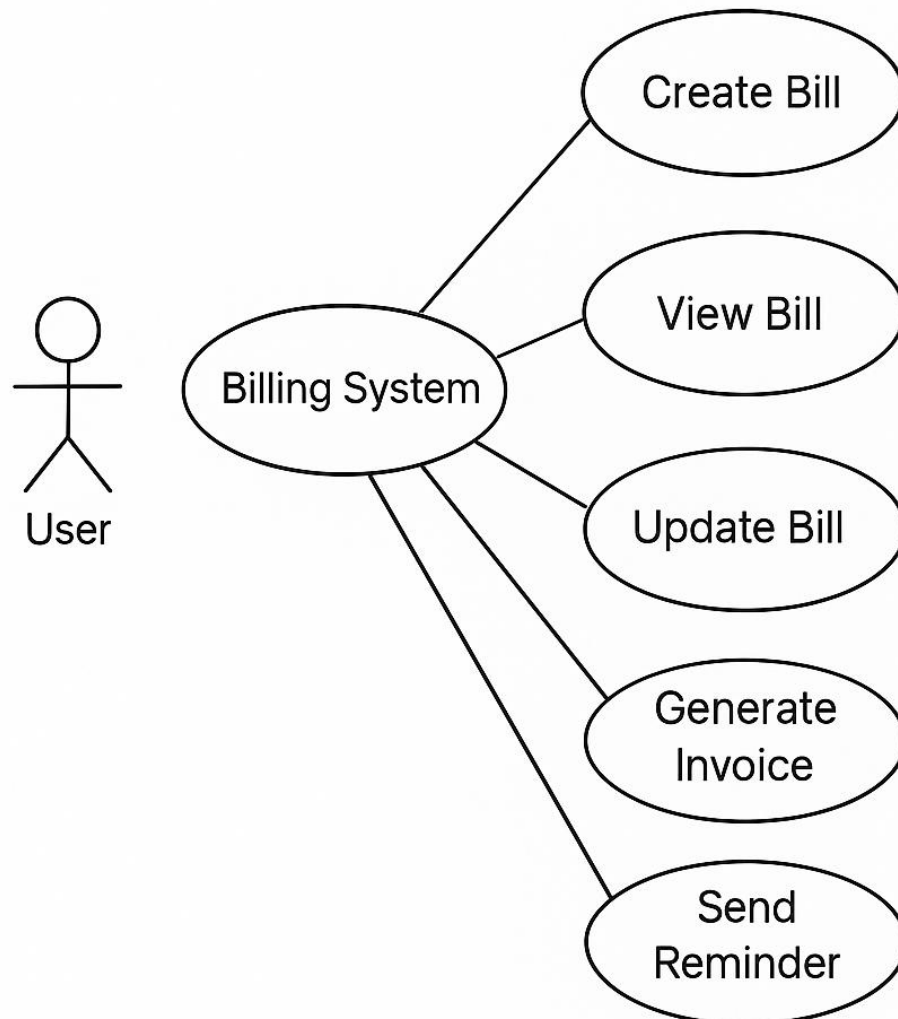
UML Class Diagram For Billing System.**5.8 UML Class Diagram for Billing System**

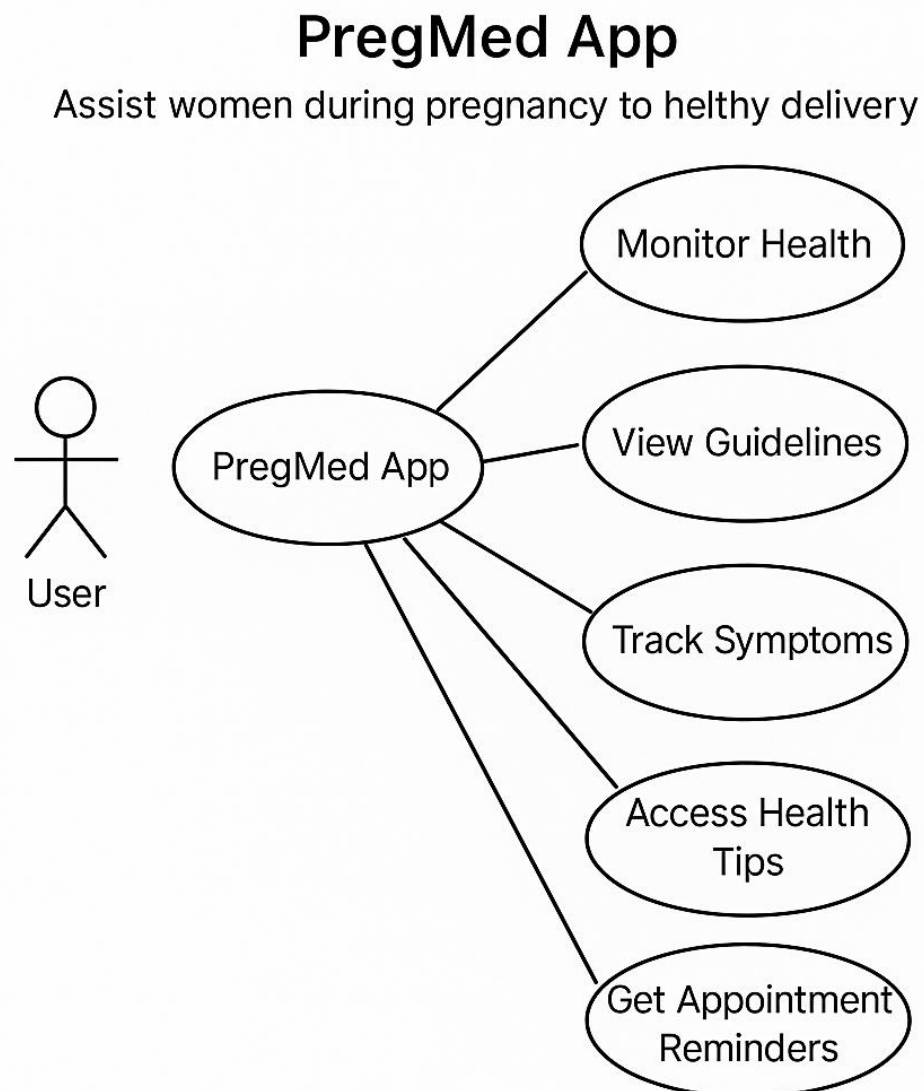
UML Class Diagram For Pregmed App.



5.9 UML Class Diagram for Pregmed App

Use-Case Diagram for OTT Platform**5.10 Use-Case Diagram for OTT Platfrom**

Use-Case Diagram for Billing System.**5.11 Use-Case Diagram for Billing System**

Use-Case Diagram for Pregmed.**5.12 Use-Case Diagram for Pregmed App**

CHAPTER 6

IMPLEMENTATION

6.1 SOFTWARE

1. OTT Platform (CCNTV) – Flutter Based

Overview: The CCNTV OTT platform is developed using **Flutter** for cross-platform compatibility and Dart as the programming language. The app is built to stream content like live TV channels, movies, and series with real-time server-based content fetching. And all age group people can enjoy semessely.

Key Implementation Features:

- **Frontend:** Flutter framework with Material UI for responsive design on Android, iOS, and Android TV.
- **Backend:** Content and metadata are fetched from a real-time server via secure REST APIs.
- **Authentication:** Integrated user login and subscription management via Firebase Authentication.
- **Streaming:** Integrated video player with HLS streaming support for seamless content delivery.
- **Subscription Management:** Monthly and yearly packages are implemented with Razorpay/Stripe payment gateway (or manually integrated options).
- **Media Caching:** Ensures smooth playback by caching recent streams.
- **Notifications:** Firebase Cloud Messaging (FCM) used for updates and promotions.

Data Tables for OTT Platform.

1.User Table

1. Users Table

Stores user information.

Column Name	Data Type	Description
user_id	INT (PK)	Unique ID for each user
name	VARCHAR	Full name of the user
email	VARCHAR (UNIQUE)	User's email address
password_hash	VARCHAR	Encrypted password
phone_number	VARCHAR	Optional phone number
profile_pic_url	VARCHAR	Profile picture URL
role	ENUM (user, admin)	Role of the user
created_at	TIMESTAMP	Account creation date

2.Movies Table

2. Movies/Shows Table

Stores all OTT content details.

Column Name	Data Type	Description
content_id	INT (PK)	Unique ID
title	VARCHAR	Movie or show title
type	ENUM (movie, show)	Content type
description	TEXT	Content synopsis
genre	VARCHAR	Comma-separated genres (e.g., Drama, Sci-Fi)
release_date	DATE	Original release date
language	VARCHAR	Primary language
age_rating	VARCHAR	e.g., PG-13, R, U/A

3.Episode Table

3. Episodes Table

Only for TV shows.

Column Name	Data Type	Description
episode_id	INT (PK)	Unique ID
content_id	INT (FK)	References <code>Movies/shows.content_id</code>
season_number	INT	Season number
episode_number	INT	Episode number within season
title	VARCHAR	Episode title
description	TEXT	Episode summary
duration	INT	Duration in minutes
video_url	VARCHAR	Link to video file or stream

4.Subscription Table

Subscriptions Table

Tracks user subscriptions.

Column Name	Data Type	Description
subscription_id	INT (PK)	Unique subscription ID
user_id	INT (FK)	References <code>Users.user_id</code>
plan_type	ENUM (Basic, Standard, Premium)	Subscription level
start_date	DATE	Start of subscription
end_date	DATE	End of subscription
is_active	BOOLEAN	If subscription is active
payment_method	VARCHAR	Card, PayPal, etc.

5.Reviews & Ratings

★ Reviews & Ratings Table

Users rate and review content.

Column Name	Data Type	Description
review_id	INT (PK)	Unique ID
user_id	INT (FK)	Reviewer
content_id	INT (FK)	Movie/Show being reviewed
rating	DECIMAL(2,1)	Rating (e.g., 4.5)
review_text	TEXT	Optional written review
created_at	TIMESTAMP	Date and time of review

6.Watchlist

📁 Watchlist Table

Stores user watchlists.

Column Name	Data Type	Description
watchlist_id	INT (PK)	Unique ID
user_id	INT (FK)	User who created the watchlist
content_id	INT (FK)	Content added to watchlist
added_at	TIMESTAMP	Date added

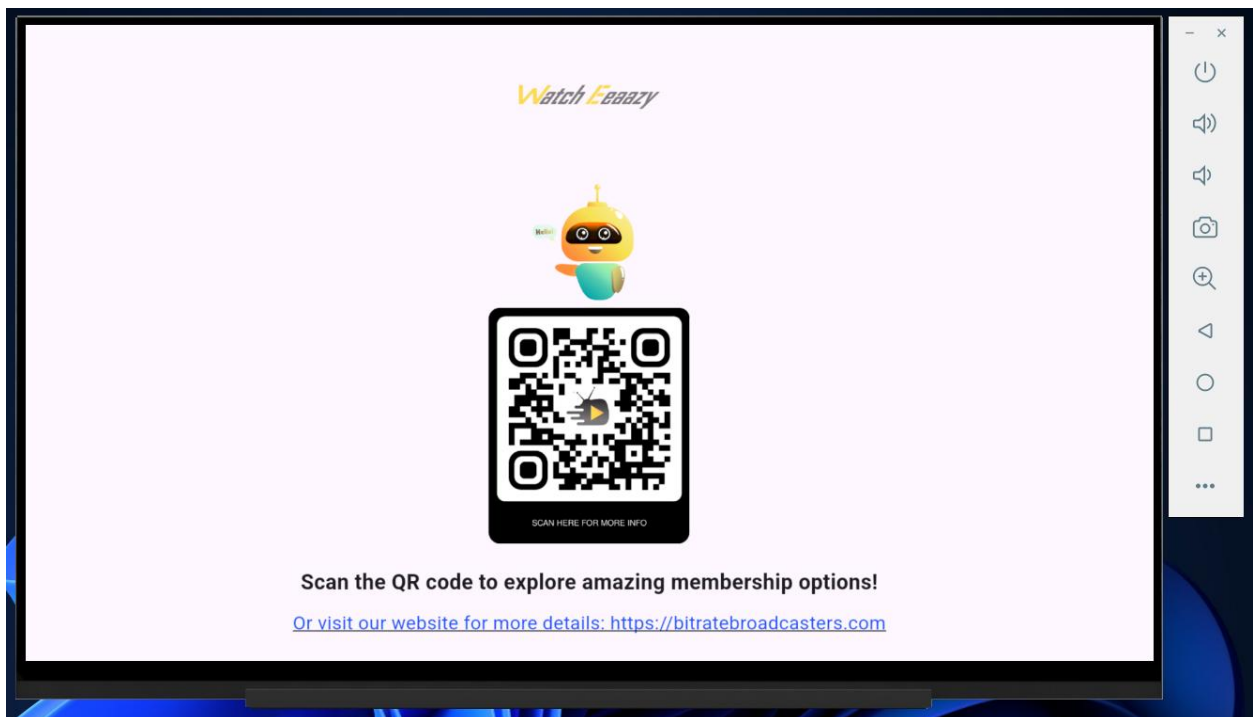
1.Splash Screen For TV



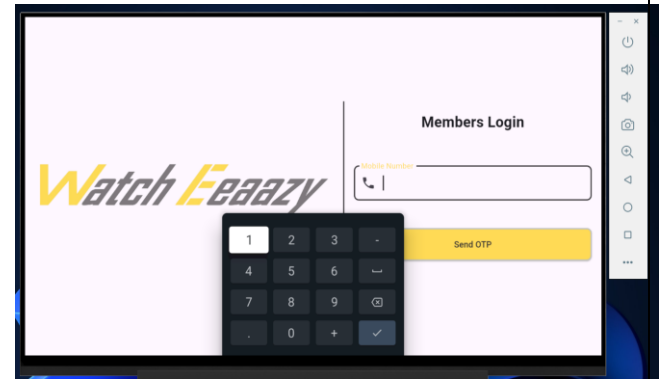
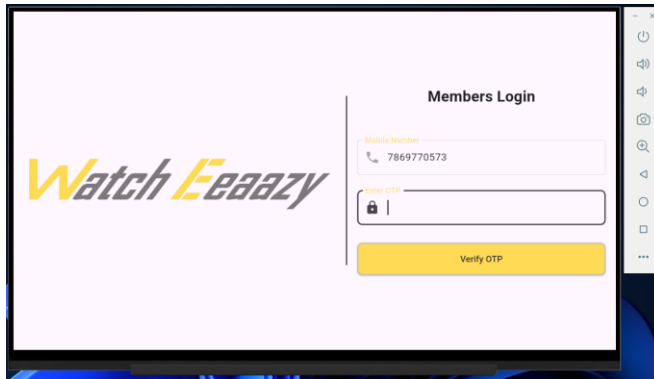
2. Onboarding Screens: -



3. Signup Screen: -



4. Login Screen: -



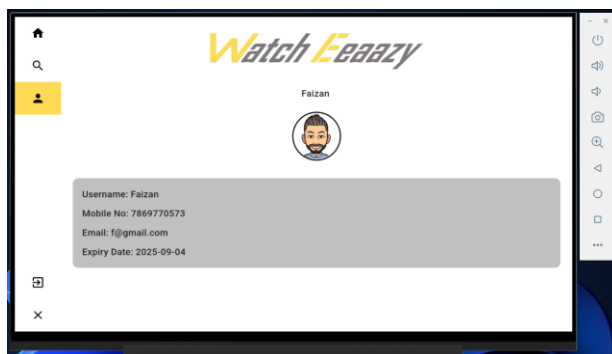
5. Home Screen: -



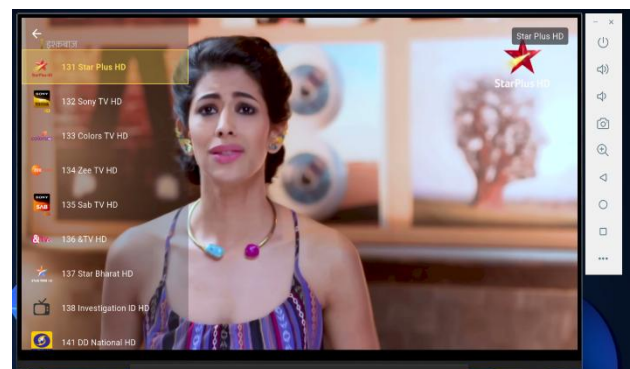
6. Search Screen: -

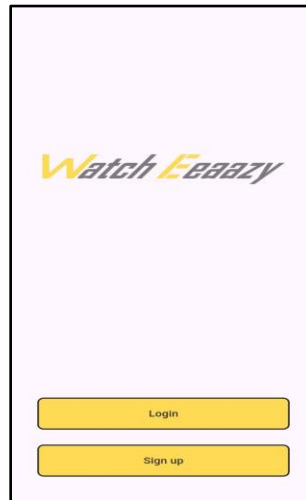
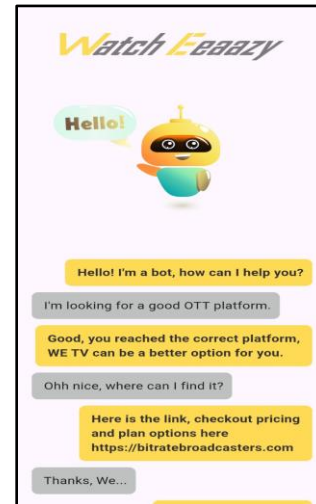
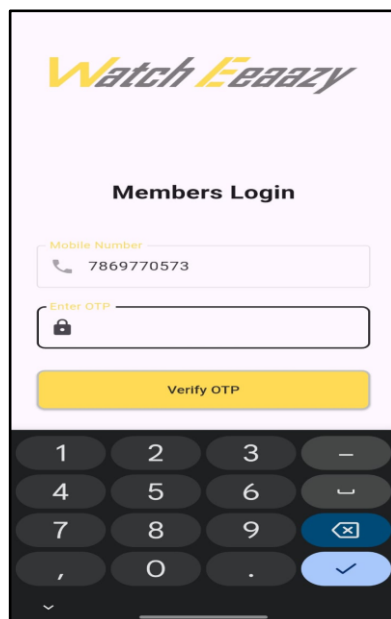
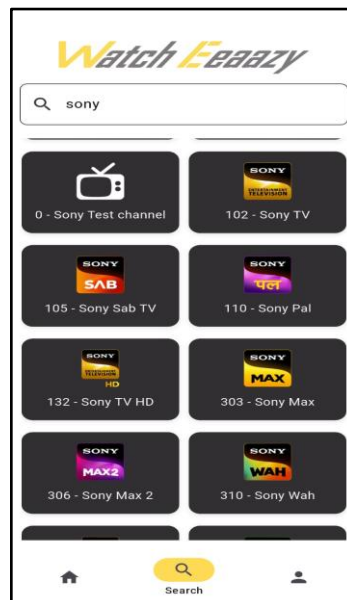
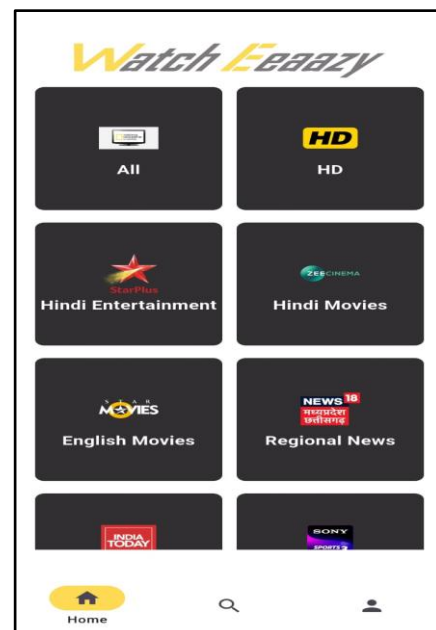


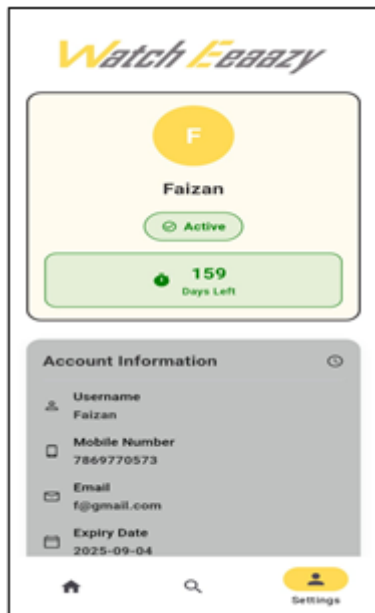
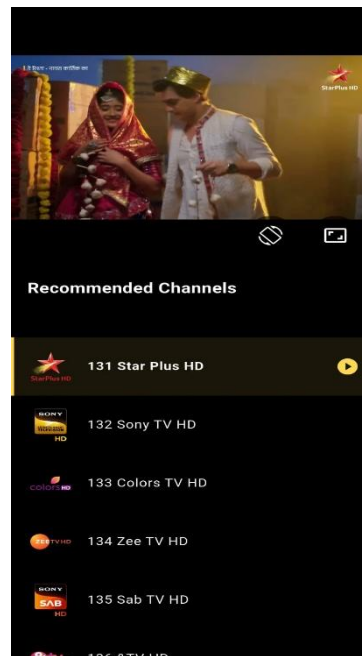
7. Profile Screen: -



8. Video Player Screen: -



Mobile Layout: -**1. Splash Screen: -****2. Onboarding Screen: -****3. Signup Screen: -****4. Login Screen: -****5. Search Screen: -****6. Home Screen: -**

7. Profile Screen: -**8. Video Player Screen: -****9. Screen Rotation: -**

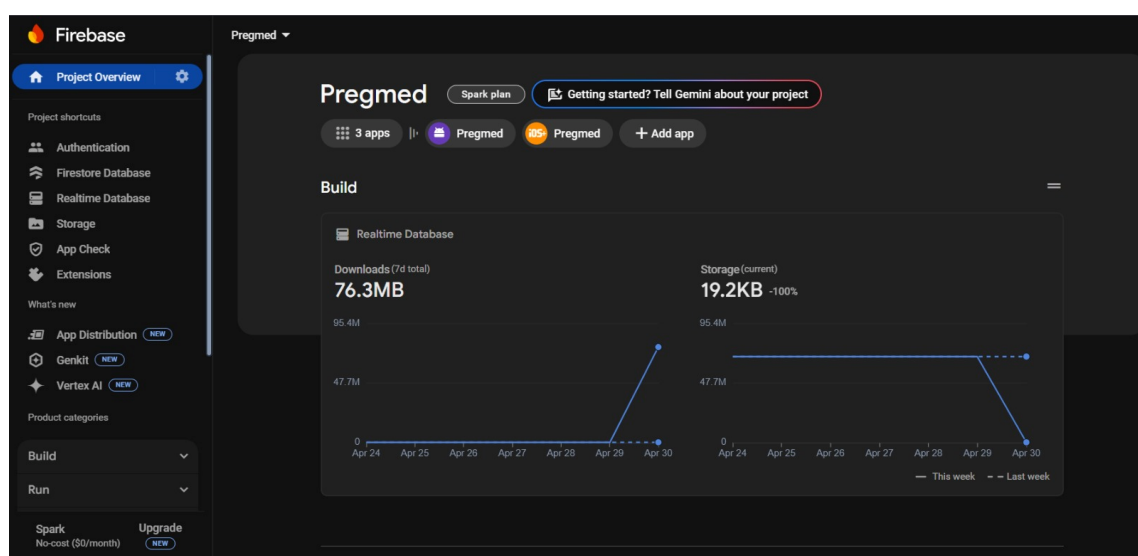
2. PregMed App – Flutter with Firebase

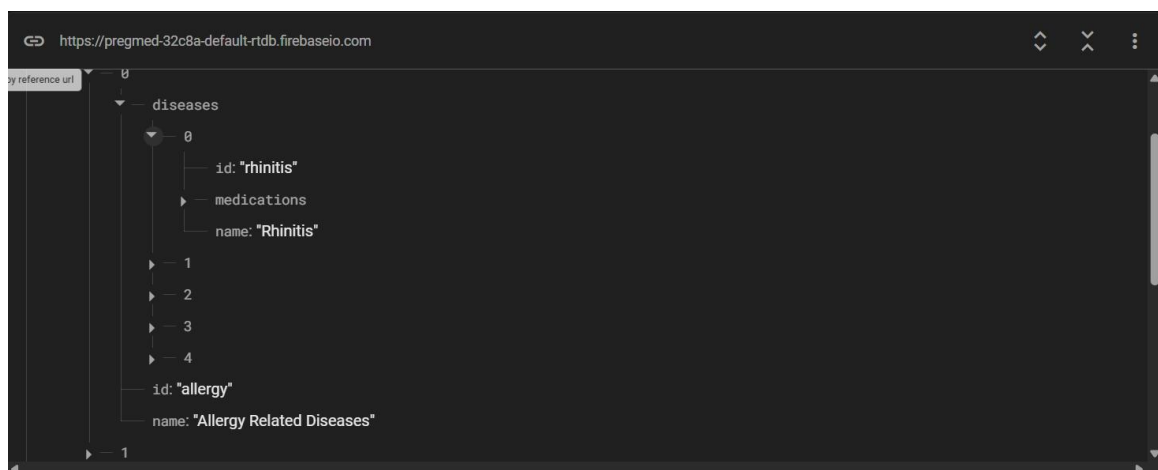
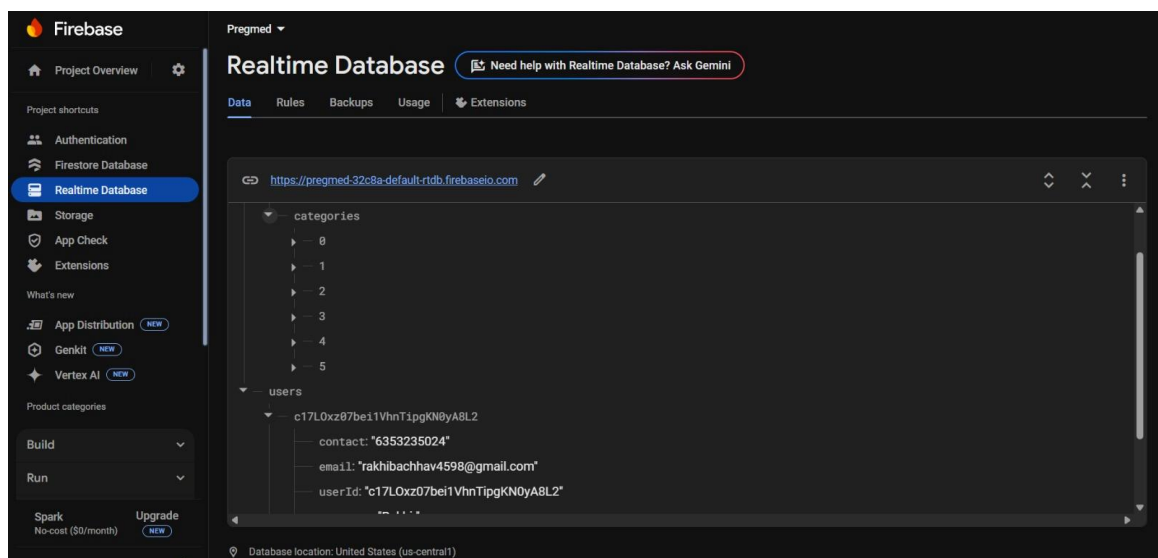
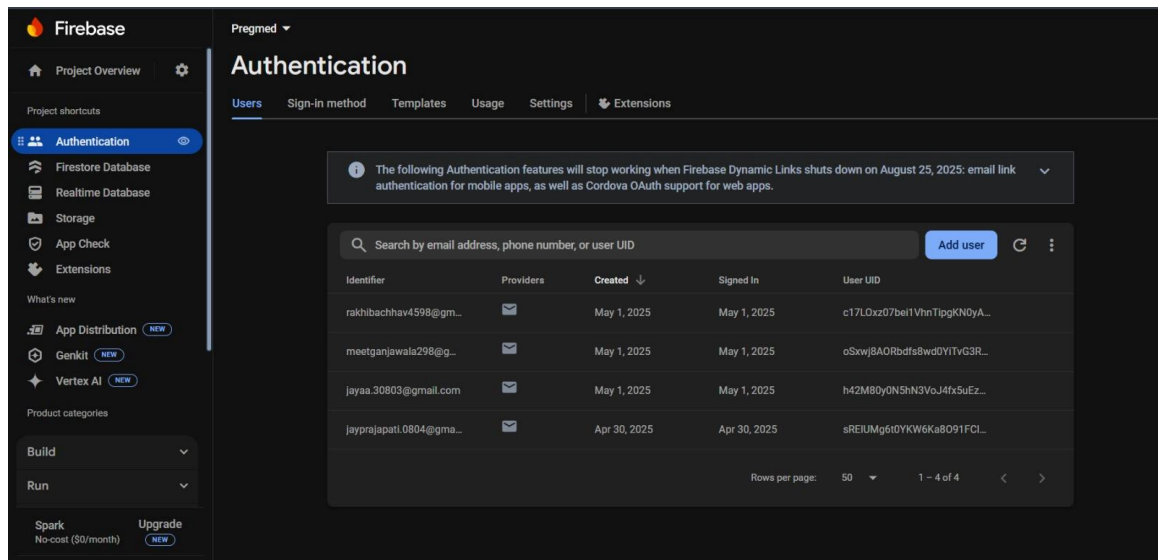
Overview: PregMed is a mobile health-tracking app designed for expecting mothers. Built with Flutter, it integrates Firebase to manage real-time data storage, user tracking, notifications, and secure login.

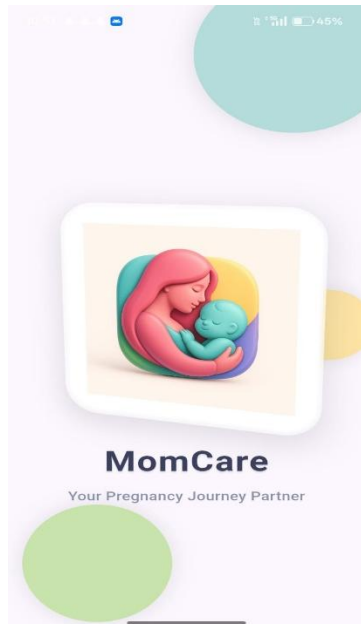
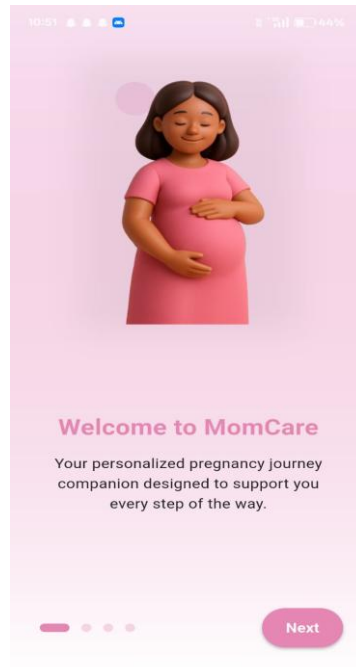
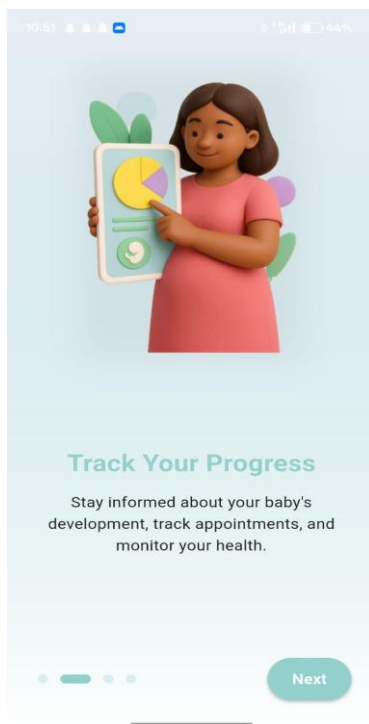
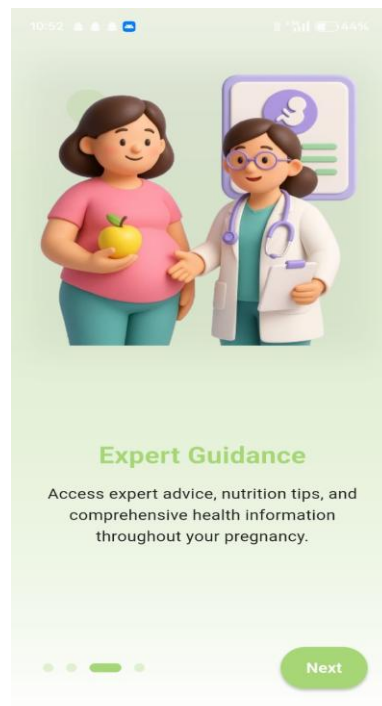
Key Implementation Features:

- **Frontend:** Built using Flutter with clean and accessible UI for pregnant users, including support for larger font sizes and intuitive navigation.
- **Backend:** Firebase (Realtime Database / Firestore) stores all user data like appointments, health logs, and medical resources.
- **Authentication:** Firebase Authentication supports email/password login and secure data access.
- **Features:**
 - Health tips and personalized suggestions based on pregnancy week.
 - Appointment reminders.
 - Health logs (weight, BP, etc.) with graphical insights.
 - Push notifications for medicine reminders and doctor follow-ups.
- **Data Sync:** Real-time syncing across multiple devices for the same user profile.

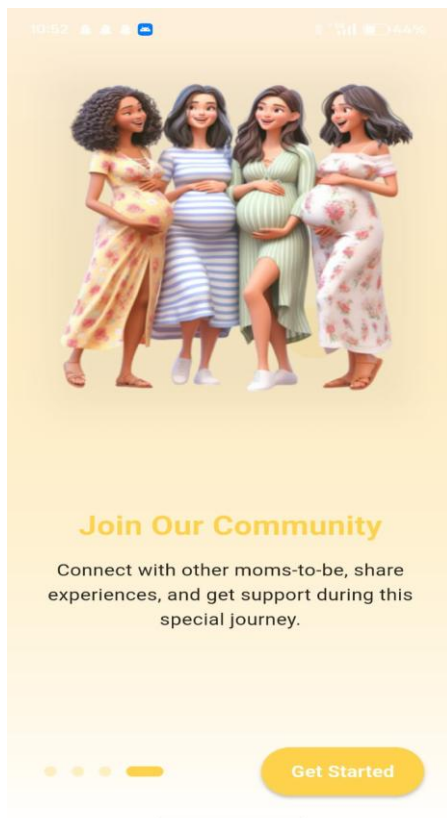
Data Tables For Pregmed App.



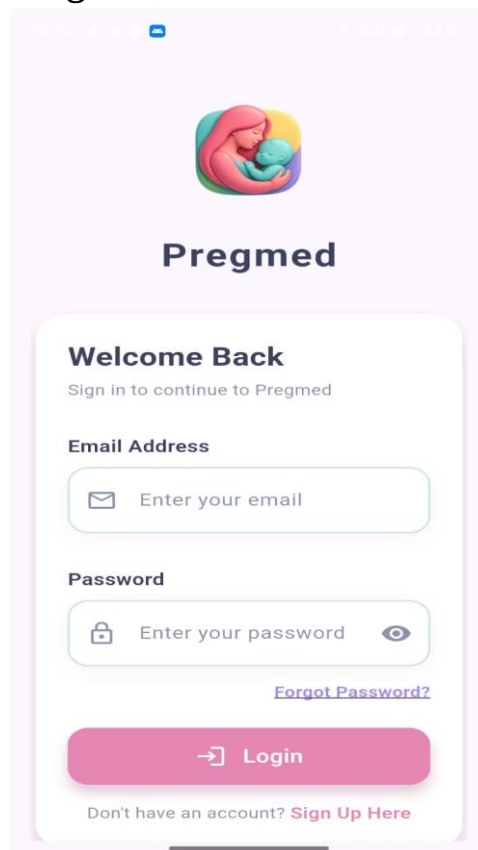


Frontend UI :-**1.Splash Screen****2.onboarding Screen****3.onboarding Screen****4. onboarding Screen**

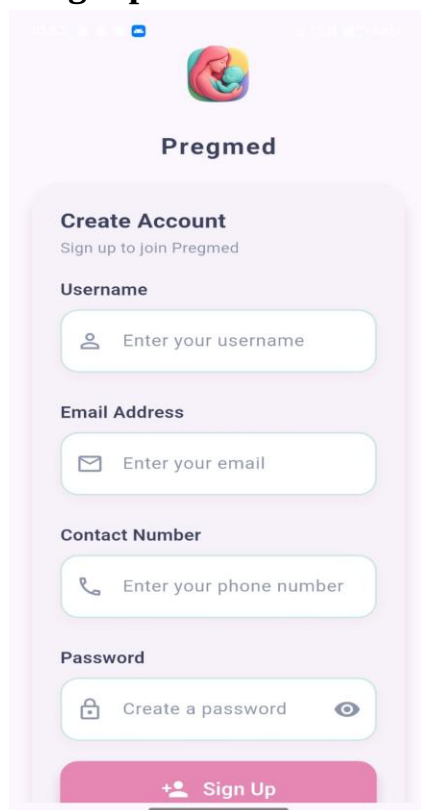
5. Onboarding Screen



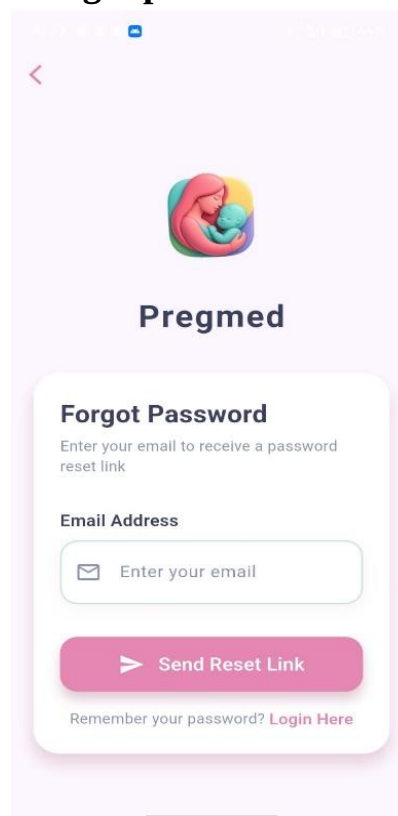
6. Login screen



7. Signup Screen



8. Signup Screen



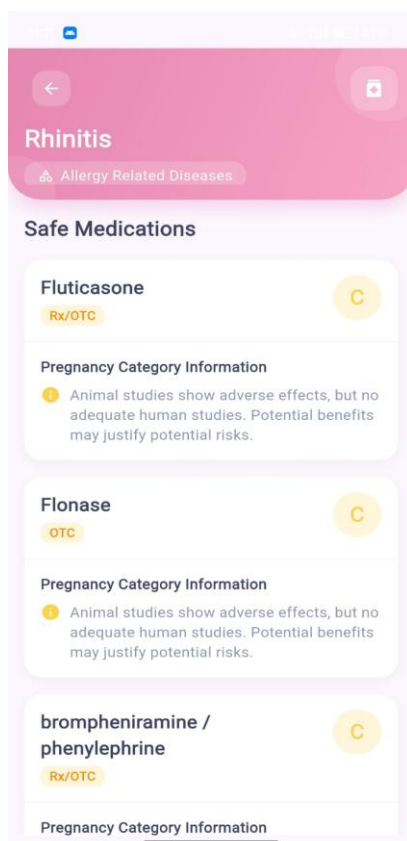
9.Home Screen



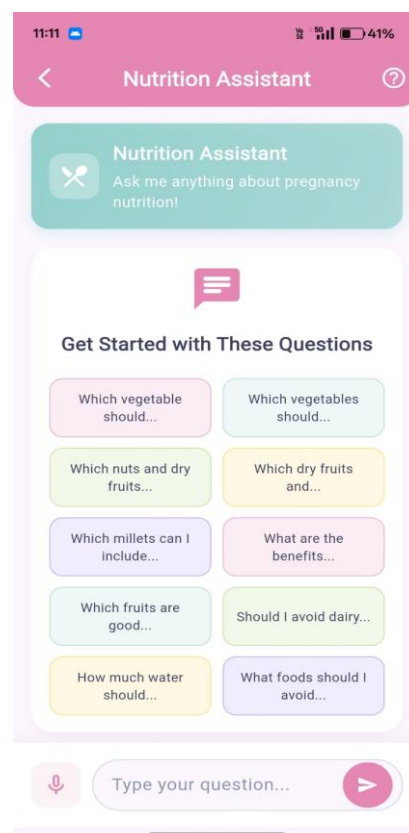
10.Medicine Screen



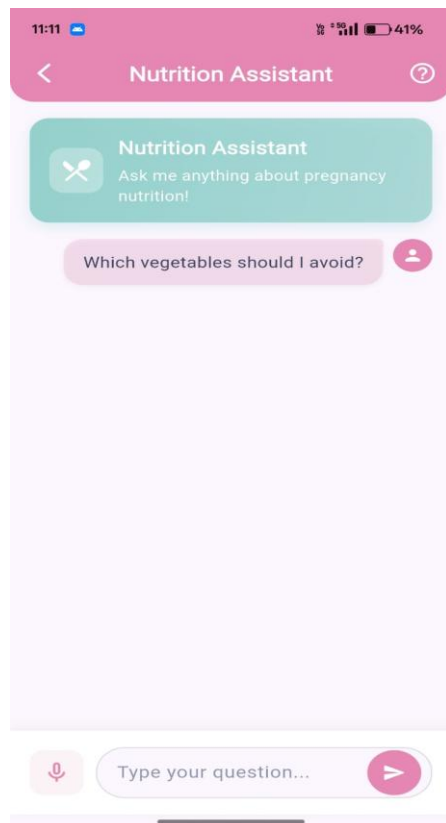
11. Medicine Details Screen



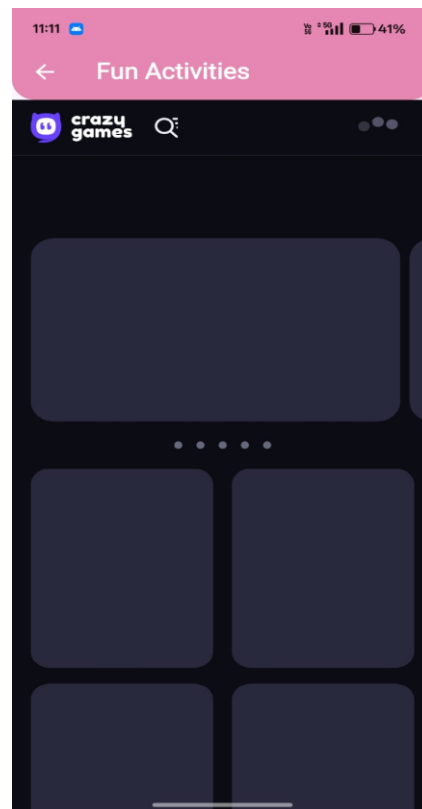
12.Nutration chat bot



13. chat bot

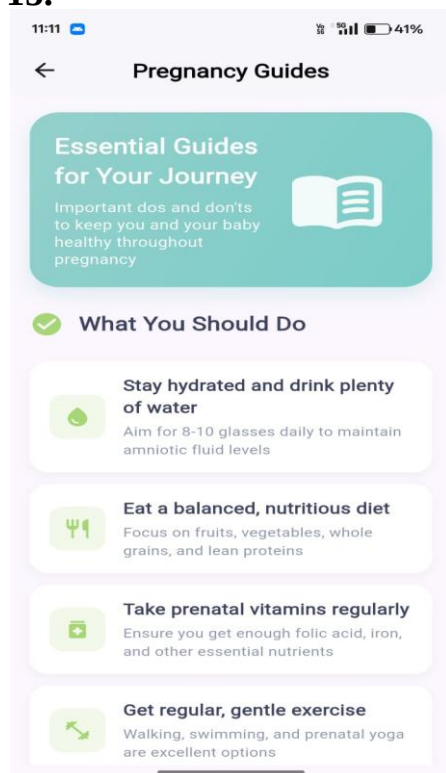


14. Fun Activities

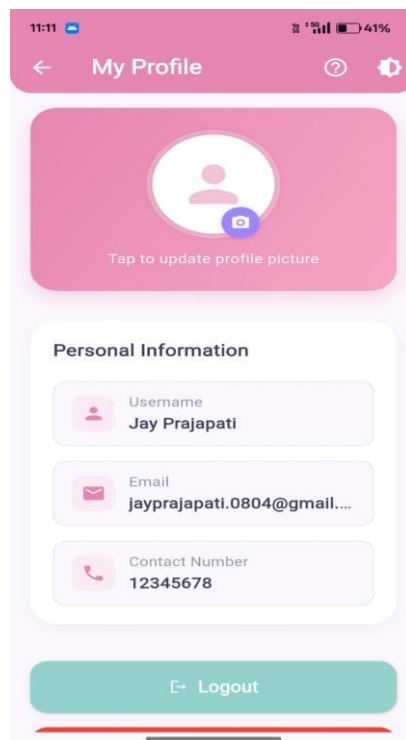


15. Guides

15.



16. Profile Screen



3. Billing System – React with PHP & MySQL

Overview: This web-based billing system is designed to generate professional invoices, maintain item masters, and handle tax calculations. The system is built using React.js for the frontend and PHP-MySQL for the backend.

Key Implementation Features:

- Frontend: Built with React.js and Tailwind CSS for a clean, responsive UI with components for forms, dashboards, and previews.
- Backend: PHP handles API requests, invoice generation, user authentication, and database CRUD operations.
- Database: MySQL stores user records, items, invoice history, and profile settings.
- Features:
 - Login and Signup with JWT-based authentication.
 - Items Master: Add, update, delete product data (with item codes and tax rates).
 - Billing Form: Add client details, line items, tax slabs, and preview before generating invoice.
 - Invoice Preview: Downloadable invoice in PDF format using jsPDF or similar library.
 - Admin Dashboard: Overview of total bills, clients, and revenue.
 - Secure session management and error handling for reliability.

DataBase Photos :-

Table structure Relation view

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐	2 user_id	int(11)			No	None			Change Drop More
☐	3 firebase_session_id	varchar(255)	utf8mb4_general_ci		No	None			Change Drop More
☐	4 ip_address	varchar(45)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	5 user_agent	text	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	6 login_time	datetime			Yes	current_timestamp()			Change Drop More
☐	7 last_activity	datetime			Yes	current_timestamp()			Change Drop More
☐	8 expires_at	datetime			Yes	NULL			Change Drop More

☐ Check all With selected: Browse Change Drop Primary Unique Index Spatial Fulltext Add to central columns
 Remove from central columns

Table structure | Relation view

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	int(11)			No	None		AUTO_INCREMENT	
☐	2 user_id	int(11)			No	None			
☐	3 activity_type	enum('login', 'logout', 'action', 'signup')	utf8mb4_general_ci		No	None			
☐	4 ip_address	varchar(45)	utf8mb4_general_ci		Yes	NULL			
☐	5 user_agent	text	utf8mb4_general_ci		Yes	NULL			
☐	6 created_at	timestamp			No	current_timestamp()			

☐ Check all With selected:

Table structure | Relation view

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	int(11)			No	None		AUTO_INCREMENT	
☐	2 firebase_uid	varchar(128)	utf8mb4_general_ci		No	None			
☐	3 username	varchar(100)	utf8mb4_general_ci		No	None			
☐	4 email	varchar(100)	utf8mb4_general_ci		No	None			
☐	5 mobile	varchar(15)	utf8mb4_general_ci		No	None			
☐	6 created_at	timestamp			No	current_timestamp()			
☐	7 updated_at	timestamp			No	current_timestamp()		ON UPDATE CURRENT_TIMESTAMP()	

☐ Check all With selected:

Table structure | Relation view

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	int(11)			No	None		AUTO_INCREMENT	
☐	2 rate	decimal(5,2)			No	None			
☐	3 description	varchar(100)	utf8mb4_general_ci		Yes	NULL			

☐ Check all With selected:

Table structure | Relation view

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	int(11)			No	None		AUTO_INCREMENT	
☐	2 item_code	varchar(50)	utf8mb4_general_ci		No	None			
☐	3 item_name	varchar(255)	utf8mb4_general_ci		No	None			
☐	4 price	decimal(10,2)			No	None			
☐	5 user_id	int(11)			No	None			
☐	6 created_at	timestamp			No	current_timestamp()			
☐	7 updated_at	timestamp			No	current_timestamp()		ON UPDATE CURRENT_TIMESTAMP()	

☐ Check all With selected:

Table structure | Relation view

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	int(11)			No	None		AUTO_INCREMENT	
☐	2 invoice_id	int(11)			No	None			
☐	3 item_id	int(11)			No	None			
☐	4 hsn_sac	varchar(50)	utf8mb4_general_ci		Yes	NULL			
☐	5 quantity	int(11)			No	None			
☐	6 rate	decimal(10,2)			No	None			
☐	7 amount	decimal(12,2)			No	None			
☐	8 cgst_rate	decimal(5,2)			No	None			
☐	9 sgst_rate	decimal(5,2)			No	None			
☐	10 cgst_amount	decimal(10,2)			No	None			
☐	11 sgst_amount	decimal(10,2)			No	None			
☐	12 total_amount	decimal(12,2)			No	None			
☐	13 created_at	timestamp			No	current_timestamp()			
☐	14 updated_at	timestamp			No	current_timestamp()		ON UPDATE CURRENT_TIMESTAMP()	

☐ Check all With selected:

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐	2 invoice_no	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More
☐	3 invoice_date	date			No	None			Change Drop More
☐	4 chalan_no	varchar(50)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	5 client_id	int(11)			No	None			Change Drop More
☐	6 user_id	int(11)			No	None			Change Drop More
☐	7 subtotal	decimal(12,2)			No	None			Change Drop More
☐	8 tax_amount	decimal(12,2)			No	None			Change Drop More
☐	9 total_amount	decimal(12,2)			No	None			Change Drop More
☐	10 created_at	timestamp			No	current_timestamp()			Change Drop More
☐	11 updated_at	timestamp			No	current_timestamp()		ON UPDATE CURRENT_TIMESTAMP()	Change Drop More

☐ Check all With selected: [Browse](#) [Change](#) [Drop](#) [Primary](#) [Unique](#) [Index](#) [Spatial](#) [Fulltext](#) [Add to central columns](#)
[Remove from central columns](#)

UI:-

1.Loading Screen :-



2.Signup Screen



Sign Up

Username

Jay Prajapati

Email Address

jayprajapatiip@gmail.com

Contact Number

1234567890

Password

Password must be at least 6 characters

Confirm Password

Create Account →

Already have an account? [Sign In](#)

3.Dashboard Screen

Total Engineering

Home

Data Entry

Invoice Generator

Download Invoice

Profile

Logout

Total Invoices

156

Pending Payments

12

Total Revenue

\$24,500

New Clients

8

INVOICE ID	CLIENT	AMOUNT	STATUS	DATE	ACTIONS
INV-001	Acme Corp	\$1,200	Paid	2025-04-18	View Edit
INV-002	Globex Inc	\$850	Pending	2025-04-15	View Edit
INV-003	Stark Industries	\$3,200	Paid	2025-04-10	View Edit
INV-004	Wayne Enterprises	\$1,800	Overdue	2025-04-05	View Edit
INV-005	Cyberdyne Systems	\$950	Pending	2025-04-01	View Edit

3.Item entry Screen

Total Engineering

Home

Data Entry

Invoice Generator

Download Invoice

Profile

Logout

Jay Prajapati
jay@example.com

Data Entry

Item Code

Enter product code (e.g., PRD001)

Item Name

Enter product name (e.g., Steel Pipe)

Price

Enter unit price (e.g., 29.99)

Add Item

4.invoice entry Screen

Total Engineering

Home

Data Entry

Invoice Generator

Download Invoice

Profile

Logout

Jay Prajapati
jay@example.com

Purchase Order Number

Purchase Item Details

No.	Item Code	Item Name	HSN/SDC	Quantity	Rate	Subtotal	Action
1	Select Code	Select Item	HSN Code	1	0	₹0.00	

+ Add Item

Tax Details

CGST (%)

0%

SGST (%)

0%

Summary

Subtotal:

₹0.00

CGST (0%):

₹0.00

SGST (0%):

₹0.00

Total:

₹0.00

Generate Invoice

4.invoice generating Screen

Total Engineering

Home

Data Entry

Invoice Generator

Download Invoice

Profile

Logout

Mo.: 9725019650

Email: totalengineering1986@gmail.com
GSTIN: 24BVYPK4491HIZY

TOTAL ENGINEERING

B-69, Mona Park, Jambusar bypass Road, Bharuch

M/s. MERINO INDUSTRIES LTD
VADODARA DAHEJ
GSTIN: 24AAACG9156C1ZT
PO. No. 3200002365

Invoice No. 76
Date 24.01.2025
Challan No. 310
Date 24.01.2025

Sl.	Description of Goods	Quantity	Unit	Rate	Amount
01	MECHANICAL SEAL	40169320	02	1,750/-	3,500/-

Bank Detail:
HDFC BANK - BHARUCH
A/C NO.: 502009007/2858
IFSC: HDFC0000068

COST 5%

315/-

COST 5%

315/-

Rs. Four Thousand One Hundred Thirty Rupees Only

TOTAL

4,130/-

TERMS & CONDITIONS:

1. Delivery Term: 1-2 week on O F O R basis after receipt PO complete in
2. Warranty: Spare parts are not covered by any warranty
3. No liability for delayed or non permission due to force majeure & reasons beyond its control
4. Service charges for type pressing and itement will be extra as actual.

Subject to Bharuch jurisdiction

For, TOTAL ENGINEERING

Jay Prajapati
jay@ensample.com

JP

CHAPTER 7

TESTING

Testing for PregMed (Flutter + Firebase App for Pregnant Women)

1. Functional Testing

Module Testing:

- User Authentication: Test sign-up/login using email and phone authentication through Firebase. Verify OTP flow and secure session handling.
- Profile Management: Ensure pregnant users can set up and update profiles with health details like trimester, age, and doctor details.
- Health Tips & Reminders: Validate the retrieval and scheduling of health tips and daily/weekly reminders via Firebase Cloud Messaging (FCM).
- Doctor Consultation & FAQ: Test functionality for accessing FAQs and contacting doctors or support if integrated.

Integration Testing:

- Firebase Authentication & Firestore: Test seamless integration for storing and retrieving user data, notifications, and content.
- FCM Integration: Ensure push notifications are delivered correctly based on user schedule and activity.

Regression Testing:

- Ensure health records, reminders, and tips remain unaffected after app updates or bug fixes.

2. Performance Testing

- **Load Testing:** Simulate multiple concurrent users to test Firestore read/write limits and cloud function scalability.
- **Stress Testing:** Simulate high usage (e.g., all users receiving notifications at once) and verify app stability.
- **Sync & Delay Testing:** Test delay tolerance between health update submissions and dashboard sync.

3. Security Testing

- **Authentication:** Validate Firebase auth against unauthorized access and token expiry.
- **Data Privacy:** Ensure all health data is stored securely and not accessible by other users.
- **Firestore Rules Testing:** Test Firestore security rules to restrict access to authorized users only.

4. Usability Testing

- **UI Testing:** Verify that health information, alerts, and tips are easily accessible and readable.
- **Accessibility Testing:** Ensure large fonts, readable colors, and screen reader compatibility for expecting mothers.

5. Compatibility Testing

- **Cross-Device Testing:** Test on various Android & iOS devices with different screen sizes and OS versions.
- **Offline Mode:** Test partial offline support (e.g., saving reminders or viewing cached tips).

6. Data Integrity Testing

- **Health Logs:** Ensure data like weight logs, checkup dates, and preferences are stored and retrieved accurately.
- **Reminders:** Verify reminder notifications are triggered as scheduled and logged.

Testing for Billing System (React + Vite Frontend, PHP MySQL Backend)

1. Functional Testing

Module Testing:

- **User Authentication:** Test login/signup for secure access to the billing system.
- **Item Master Module:** Validate adding/editing/deleting items with unique item codes.
- **Billing Form:** Test accurate bill creation, tax calculation, and invoice generation.
- **PDF Export:** Test previewing and exporting bills to PDF.

Integration Testing:

- **Frontend-Backend Communication:** Verify seamless data flow between React frontend and PHP backend using APIs.
- **PDF Library Integration:** Test PDF export for various invoice sizes and layouts.

Regression Testing:

- After updates to billing logic, ensure existing features like invoice preview and item management still work.

2. Performance Testing

- **Load Testing:** Simulate concurrent users generating bills to test MySQL and server load handling.
- **Query Optimization:** Monitor DB response time for large datasets and pagination.

3. Security Testing

- **SQL Injection & XSS:** Test input fields against SQL injection and cross-site scripting.
- **Session Management:** Validate session timeout and secure cookie storage.
- **Role-Based Access:** If admin/user roles exist, ensure proper access controls.

4. Usability Testing

- **UI Testing:** Check for clean form validation, error handling, and intuitive layout for bill generation.
- **Responsive Layout:** Ensure the app adjusts properly on desktop, tablet, and mobile screens.

5. Compatibility Testing

- **Browser Testing:** Validate functionality on major browsers (Chrome, Edge, Firefox, Safari).
- **Device Testing:** Ensure responsive UI on different screen sizes using Chrome DevTools.

6. Data Integrity Testing

- **Billing Accuracy:** Ensure all billing entries reflect accurate tax, item rate, and totals.
- **Database Consistency:** Ensure that all actions (create, update, delete) are reflected in the MySQL database correctly.

Testing for OTT Platform (Flutter App for Content Streaming)

1. Functional Testing

Module Testing:

- User Authentication: Test email/phone-based login, password reset, and session handling.
- Content Streaming: Ensure video playback starts quickly and runs smoothly (using appropriate caching or CDNs).
- Categories & Search: Validate categorization of content (Movies, TV Shows, etc.) and search functionality.
- Subscription (If available): Test subscription management, renewals, and plan switching.

Integration Testing:

- Video Player Integration: Test all features (play/pause/seek/resume) across platforms.
- Backend APIs: Verify content metadata, thumbnails, and playback URLs are fetched and rendered properly.

Regression Testing:

- Test content access and playback stability after updates to UI or backend APIs.

2. Performance Testing

- Load Testing: Simulate multiple users streaming content concurrently to test CDN and video server load.
- Buffering & Playback: Test average video load time, buffering behavior, and quality switching (if adaptive bitrate is used).

3. Security Testing

- Content Protection: Validate that video URLs are secure and expiring links or DRM are used if needed.
- Authentication Testing: Ensure no unauthorized access to premium content.
- Data Privacy: Protect user viewing history and credentials.

4. Usability Testing

- UI Testing: Ensure seamless navigation through the homepage, categories, and video player.
- Player UX: Test playback controls, subtitles, and full-screen mode across devices.

5. Compatibility Testing

- Cross-Device Testing: Test on Android TV, smartphones, and tablets with different OS versions.
- Landscape/Portrait: Verify video and app UI work well in both orientations (if allowed).

6. Data Integrity Testing

- Watch History: Validate that watch history is recorded accurately and resume functionality works.
- User Preferences: Ensure likes, bookmarks, and category preferences persist across sessions.
- Analytics: Verify backend logs accurate data for content views, watch time, and user actions.

CONCLUSION

Over the course of my 16-week internship at **Tech Elecon**, I had the opportunity to grow both technically and professionally. Working across two major domains—**Flutter** and **React**—I was able to deepen my full-stack development skills and contribute meaningfully to real-world projects. My role as the **Flutter Interns' Head** allowed me to lead a team, mentor fellow interns, and take ownership of key project components.

During the internship, I successfully developed and delivered **four major projects**:

1. **PregMed (Flutter)**: A healthcare application designed to assist pregnant women in tracking their pregnancy journey, featuring personalized reminders and medical tips.
2. **OTT Platform (Flutter)**: A content streaming app optimized for both mobile and TV platforms, offering smooth playback and categorized content delivery.
3. **Billing Management System (React)**: A responsive web app for generating, exporting, and managing client invoices with an integrated item master module.
4. **CRM (React)**: A client relationship management system that supports lead tracking, user management, and efficient client communication workflows.

This internship not only enhanced my technical proficiency with **Flutter**, **React.js**, **Firebase**, **PHP**, and **MySQL**, but also strengthened my understanding of **team collaboration**, **agile development**, and **project ownership**.

I am proud to share that my contributions and growth during this internship led to a **full-time job offer from Tech Elecon**, where I am excited to begin the next phase of my career. This journey has laid a solid foundation for my professional development and reaffirmed my passion for building impactful digital solutions.

REFERENCES

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Description: Official website for Flutter, the open-source UI software development toolkit by Google.
2. **Firestore Documentation**
URL: <https://firebase.google.com/docs>
Description: Official documentation for Firestore services including Authentication, Firestore, and Cloud Messaging.
3. **React – A JavaScript library for building user interfaces**
URL: <https://reactjs.org>
Description: Official React documentation by Meta (formerly Facebook).
4. **Vite – Next Generation Frontend Tooling**
URL: <https://vitejs.dev>
Description: Official site for Vite, the fast build tool and development server.
5. **MySQL – The world's most popular open source database**
URL: <https://www.mysql.com>
Description: Official site for MySQL database used in backend services.
6. **PHP Manual**
URL: <https://www.php.net/manual/en/>
Description: Comprehensive manual for PHP, the server-side scripting language used in your backend.
7. **Node.js Documentation**
URL: <https://nodejs.org/en/docs/>
Description: Official Node.js documentation used for backend API development (if applicable in CRM or future enhancements).
8. **Google Fonts & Material Icons**
URL: <https://fonts.google.com>
Description: Fonts and icon resources used in UI design of Flutter and React projects.
9. **WCAG 2.1 – Web Content Accessibility Guidelines**
URL: <https://www.w3.org/WAI/standards-guidelines/wcag/>
Description: Official accessibility guidelines referred to for usability and accessibility testing